HDSF MODULE ANATOMY

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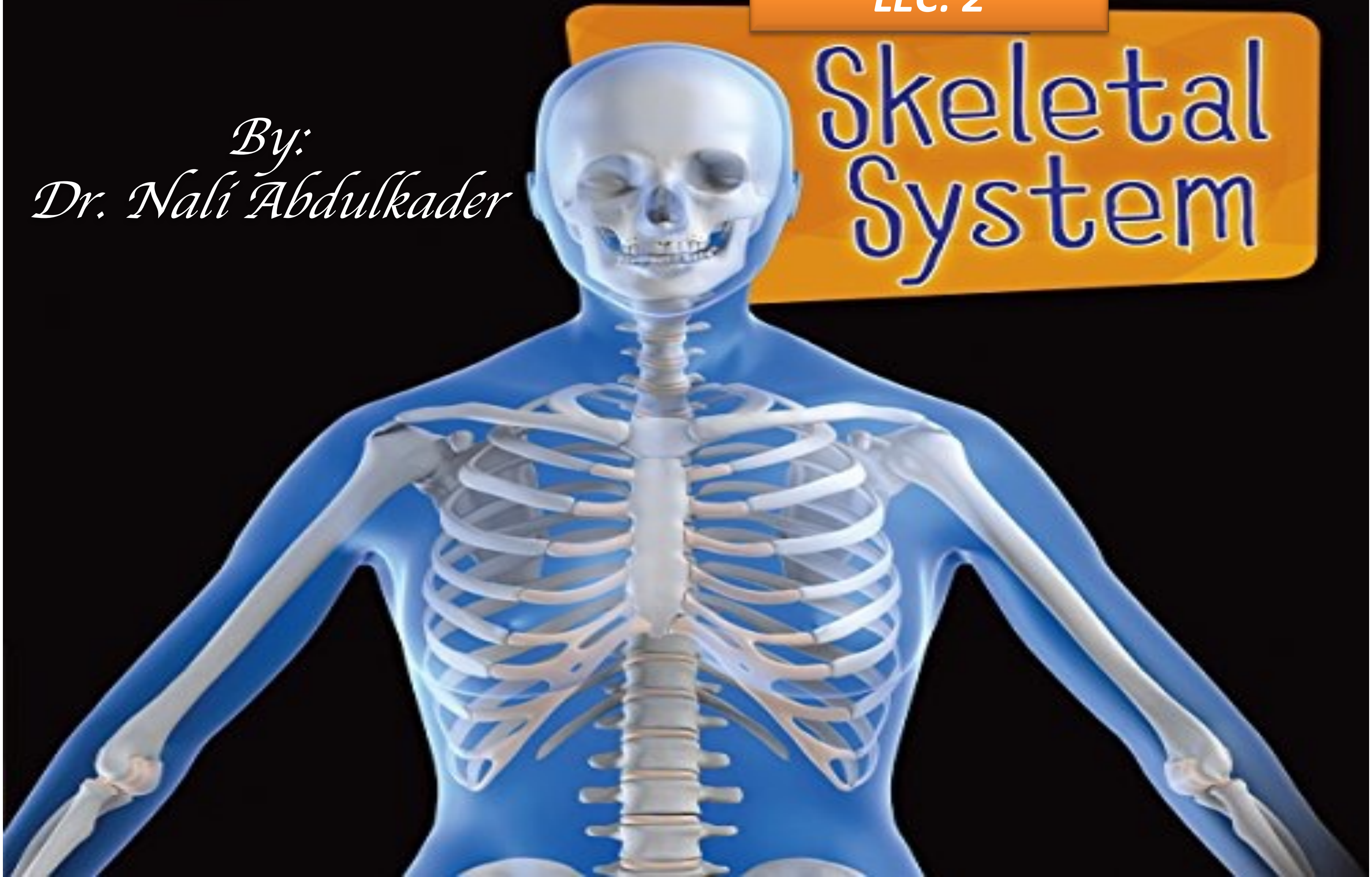
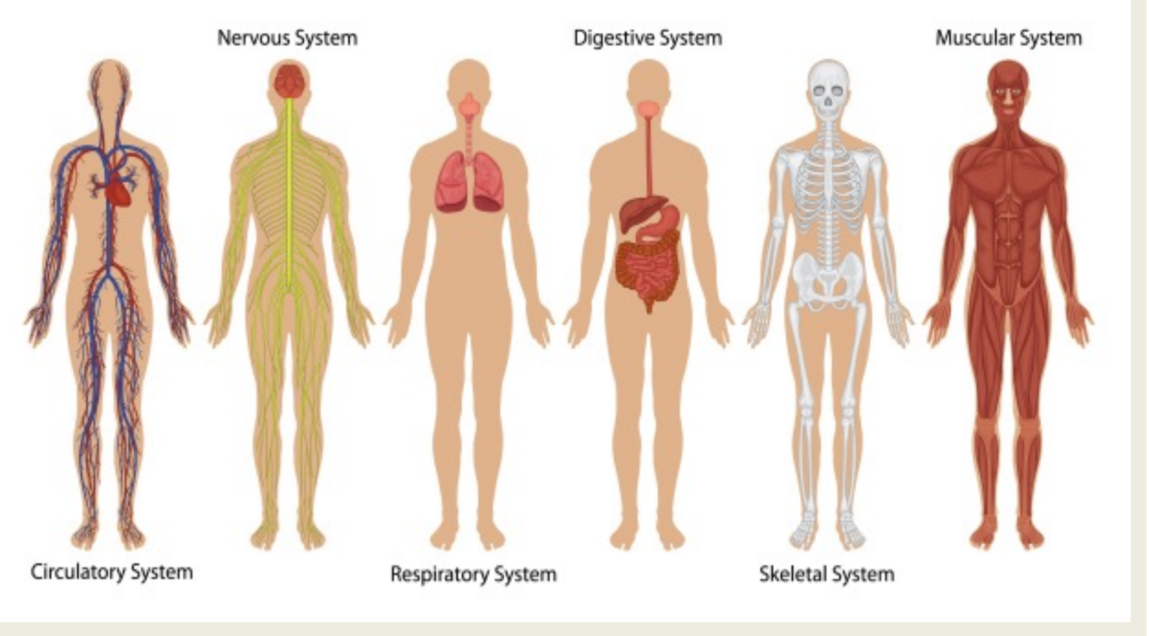


By:
Dr. Nali Abdulkader

SYSTEMIC ANATOMY

- Collective structure that work together to carry out complex function

- *Skeletal system*
- *Muscular system*
- *Respiratory system*
- *Cardiovascular system*
- *Digestive system*



Our bones form the skeletal framework of our body

206 Bones

640 Muscles

200 Joints

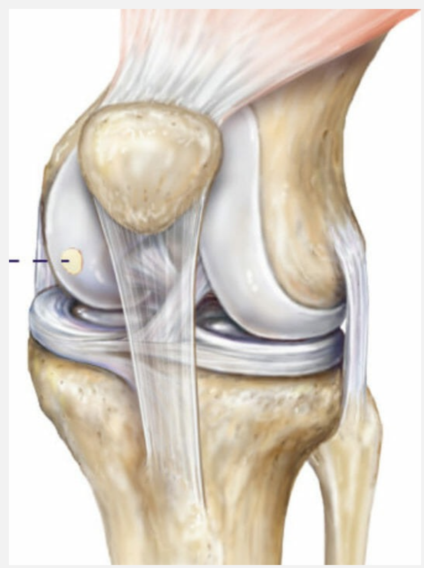
4000 Tendons

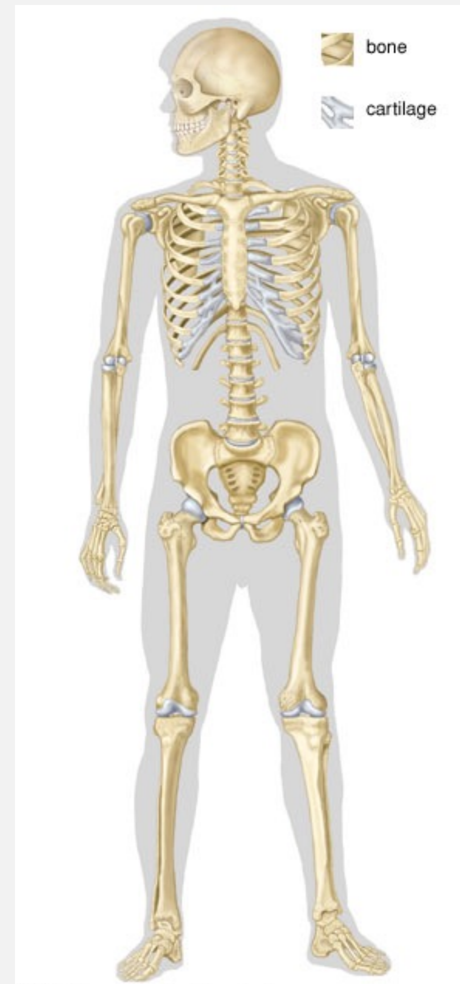
160 Bursae

900 Ligaments

Skeletal system

- Is the body system composed of bones, cartilages, ligaments and Joints
- Is makes up about 1/5th of the body weight

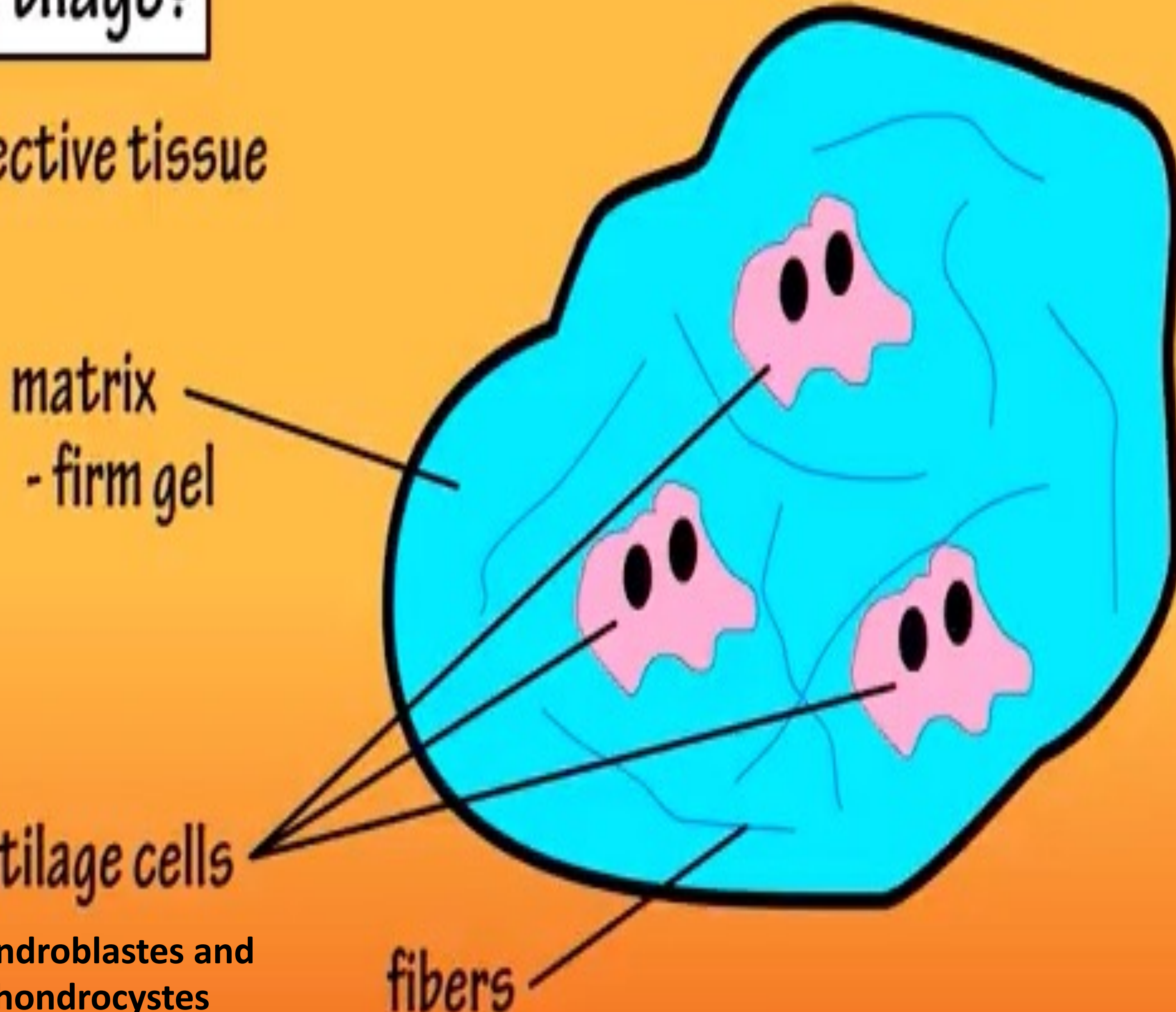





Cartilage
What Is It?

what is cartilage?

- firm connective tissue

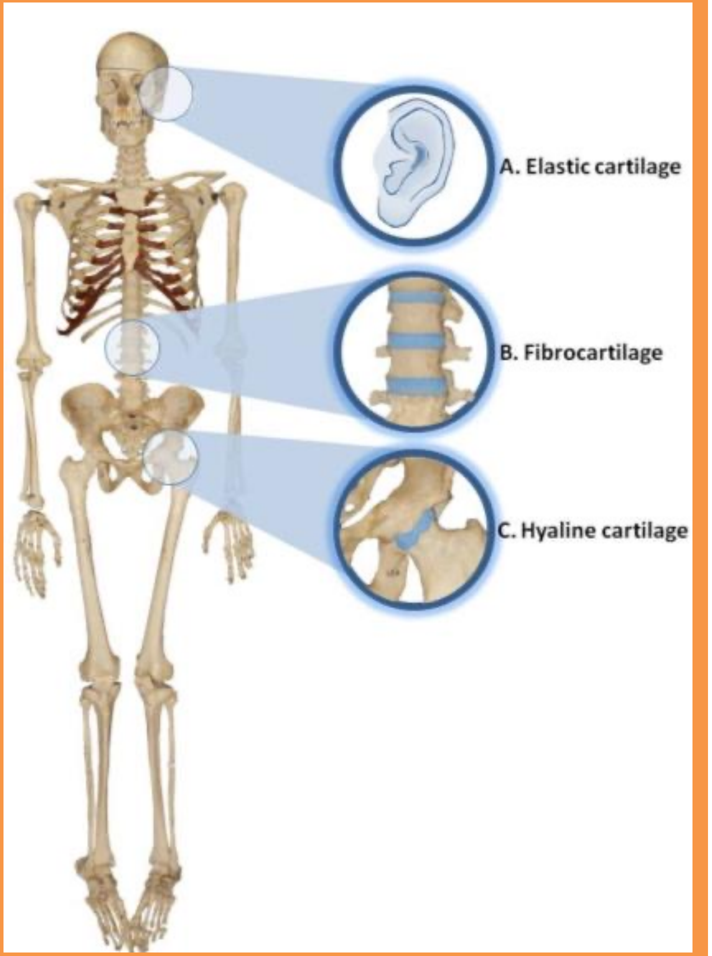


Chondroblastes and chondrocystes

Cartilage

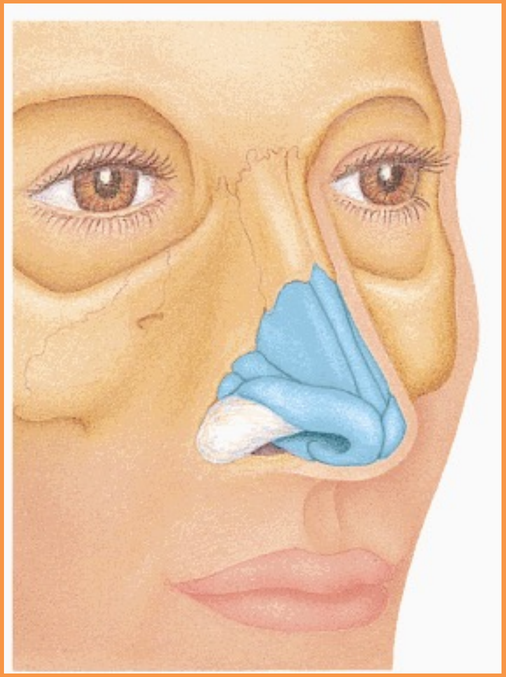
Cartilage is a special type of C.T *nonvascular* and three types

- **Hyaline Cartilage** (articular surface)
- **Firouscartilage** (discs)
- **Elastic cartilage** (auricle)

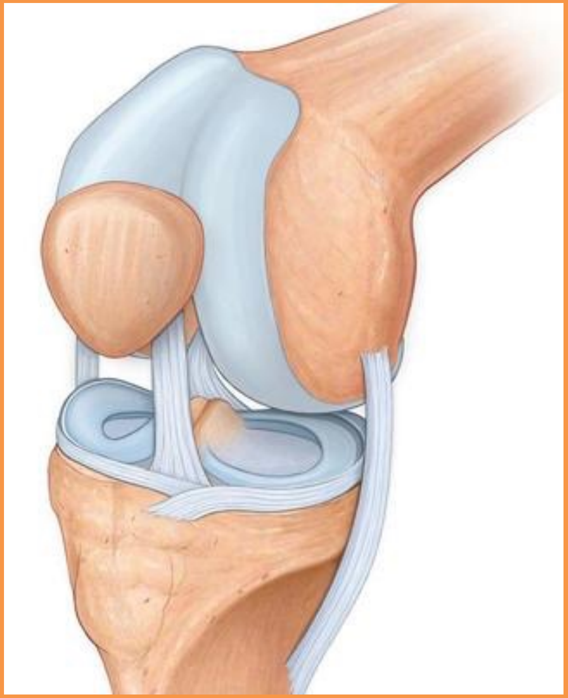


1- Hyaline cartilage

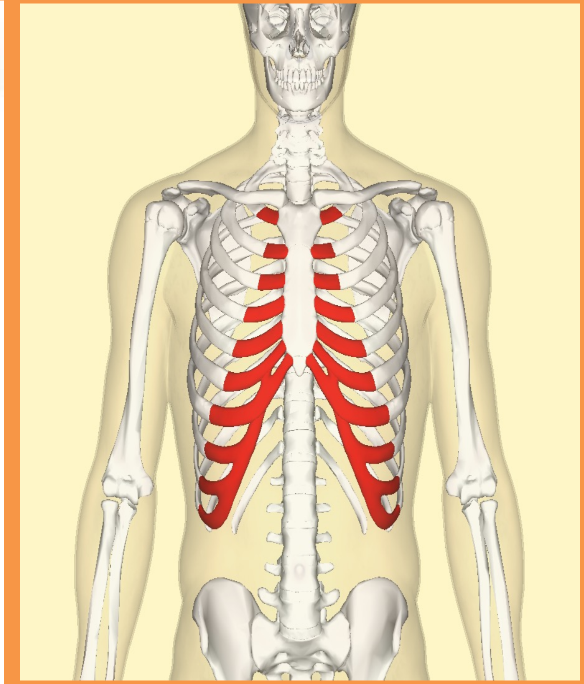
- Most common type of cartilage and is found in



Nasal septum.



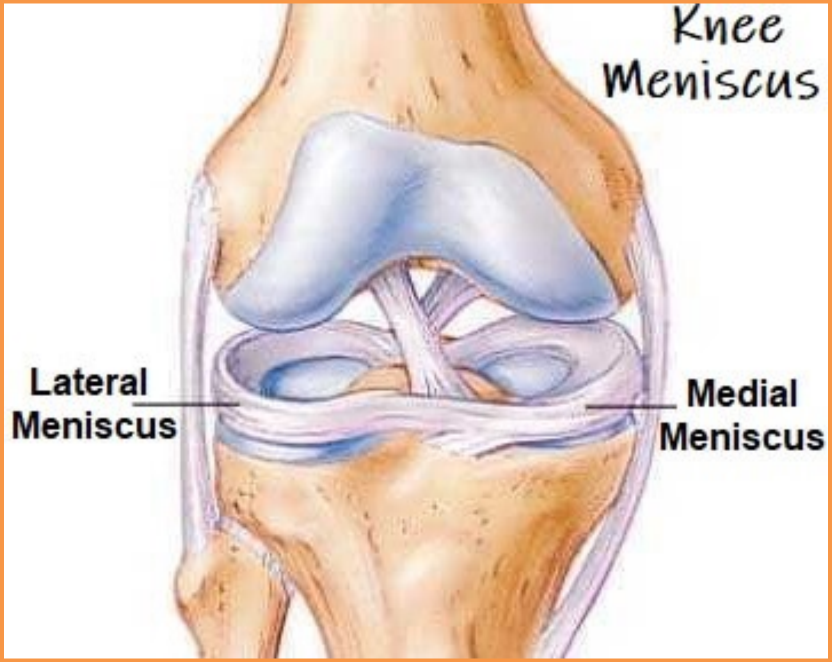
articular surface

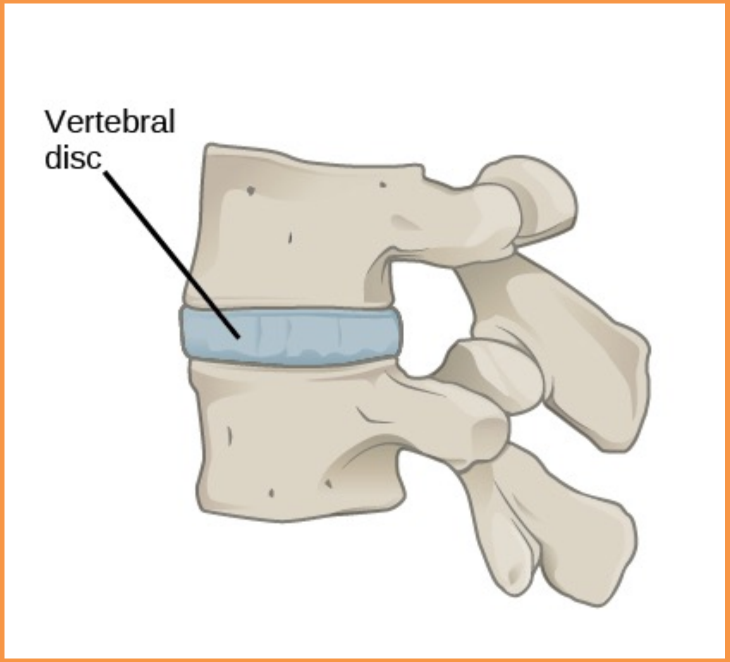


costal cartilage

2- Fibrous cartilage

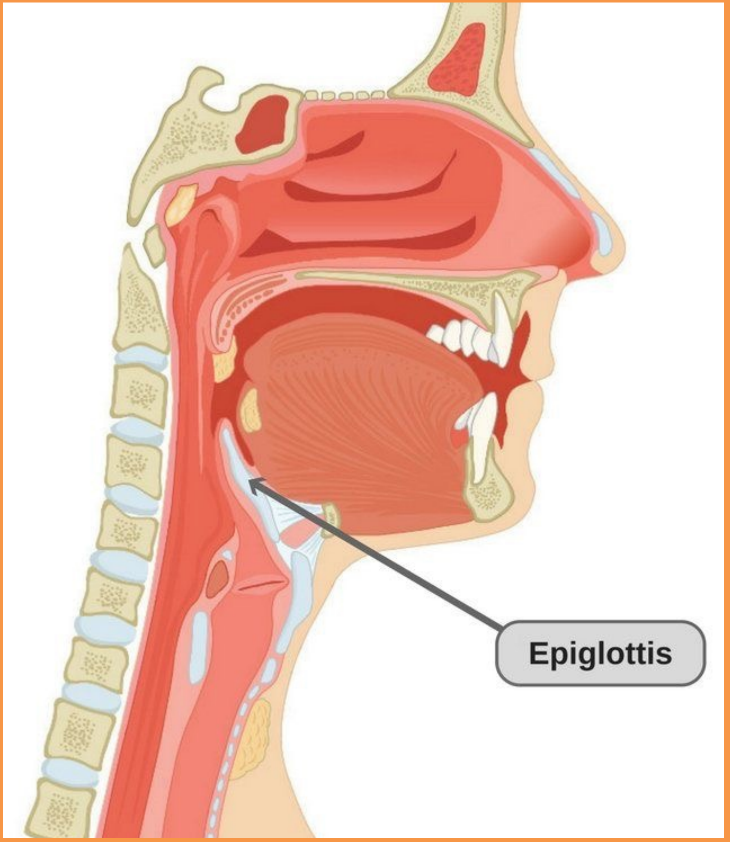
- Toughest form of cartilage and is found





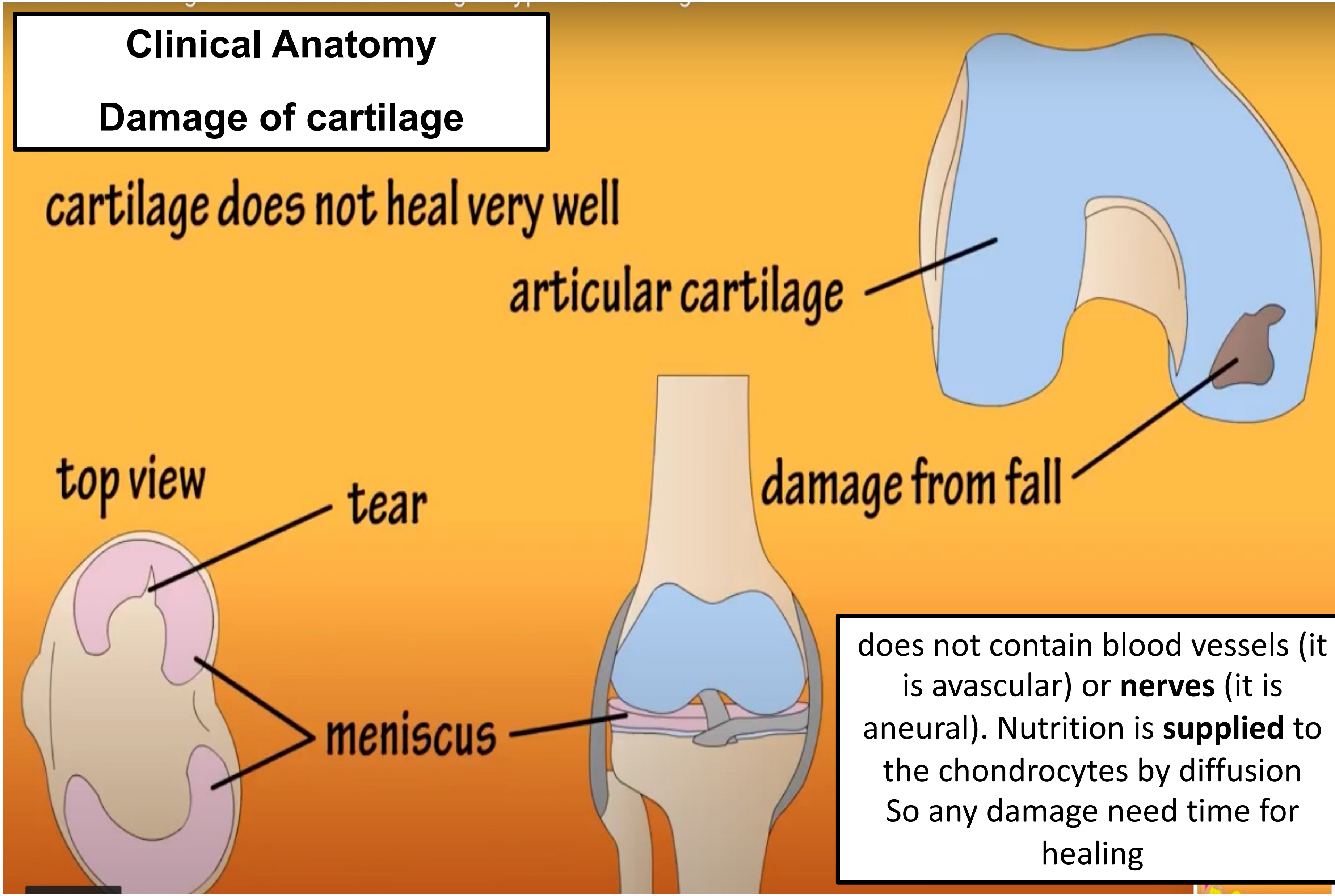
3-Elastic cartilage

- Provides firmness and elasticity and is found



Clinical Anatomy

Damage of cartilage



does not contain blood vessels (it is avascular) or **nerves** (it is aneural). Nutrition is **supplied** to the chondrocytes by diffusion
So any damage need time for healing

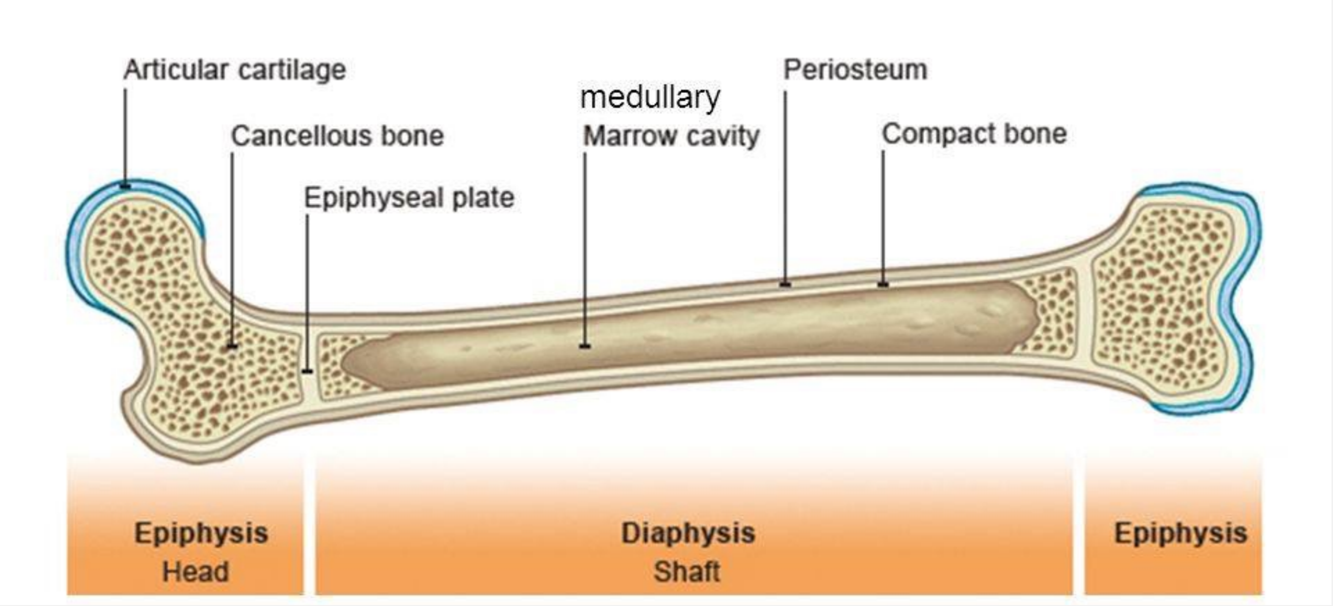
B- Bone

- **Hard calcified type of C.T its function**

1. Protects Organs
2. Offers support, protection and flexibility to organisms
3. Forms blood cells
4. Stores calcium

Bone composition

- Outer part called compact bone
- Inner part spongy bone (cancelous)
- Medullary cavity filled with bone marrow
- Covered externally by the membrane called periosteum which is supply by sensory nerve and is very sensitive for pain

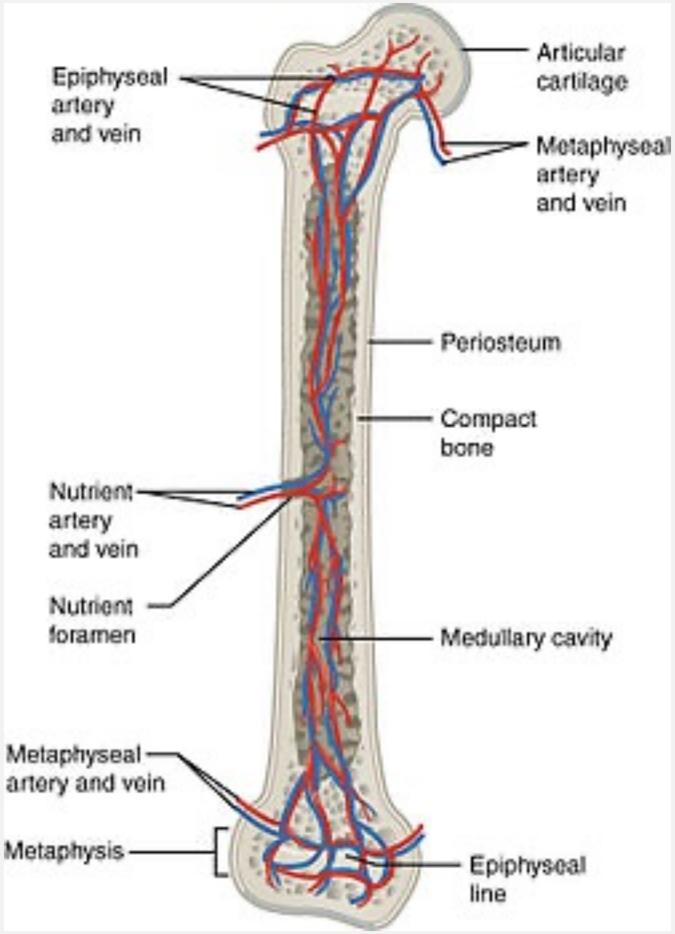


Blood Supply of Bone

1- Periosteal arteries: supply periosteum

2- Nutrient arteries enter through nutrient foramen supply diaphysis and bone marrow

3- Metaphyseal & epiphyseal arteries supply red marrow & bone tissue of metaphyses & epiphyses

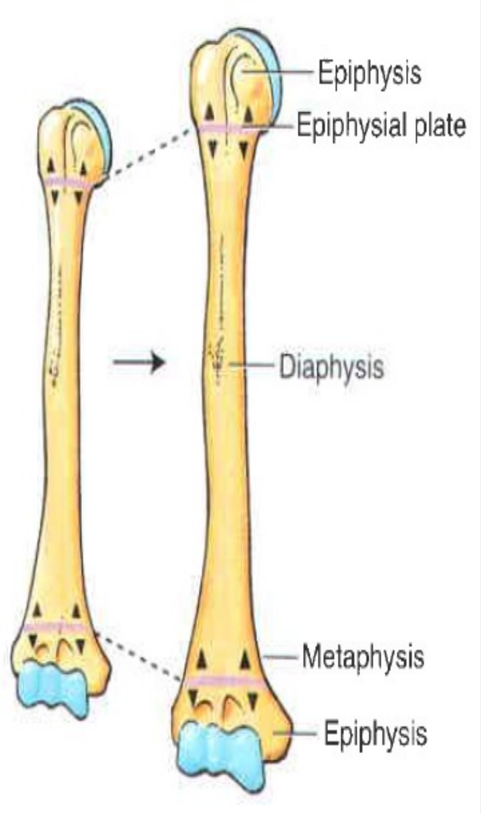


Gross types of bones

1- Long bones: have two ends and shaft. e.g. humerus and femur Each long bone consists of: diaphysis = shaft

epiphysis

metaphysis = includes epiphyseal plate in growing bones.

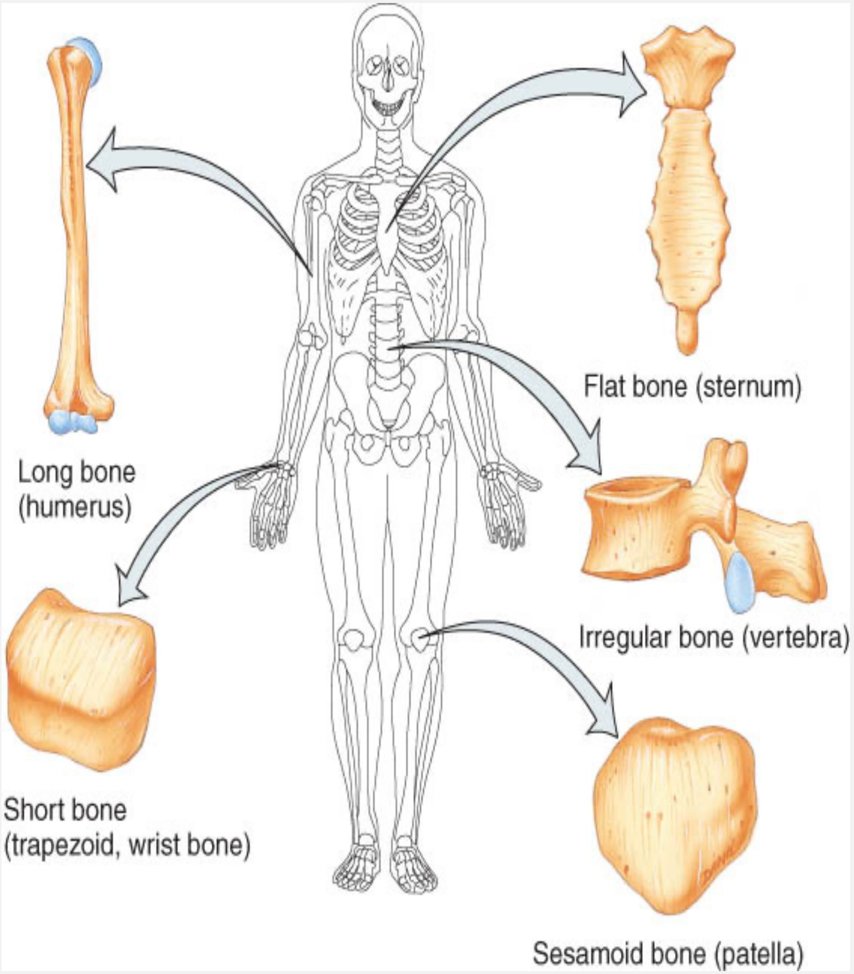


2- Short bones: have no epiphyses and shaft.

e.g. carpal & tarsal bones

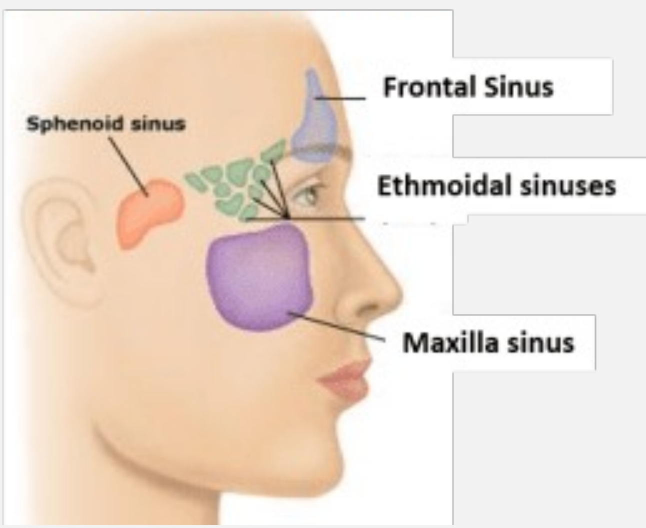
3- Flat bones: bones with two surfaces. e.g. cranial bones, scapula, ribs, sutural bones

4- Irregular bones:
bones with no definite shape.
e.g. vertebrae

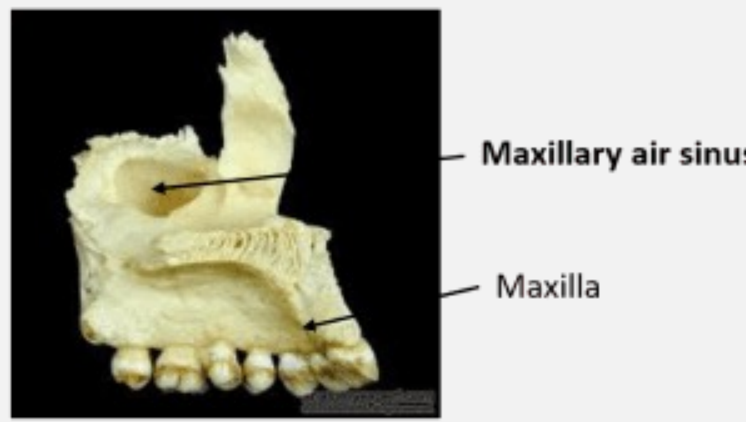


5- Sesamoid bones: associated with tendons of upper and lower limbs. Patella is the largest sesamoid bone.

6- Pneumatic bones: certain bones of skull containing air cavities, such as; frontal, sphenoid, ethmoid, mastoid part of temporal and maxillae. …….

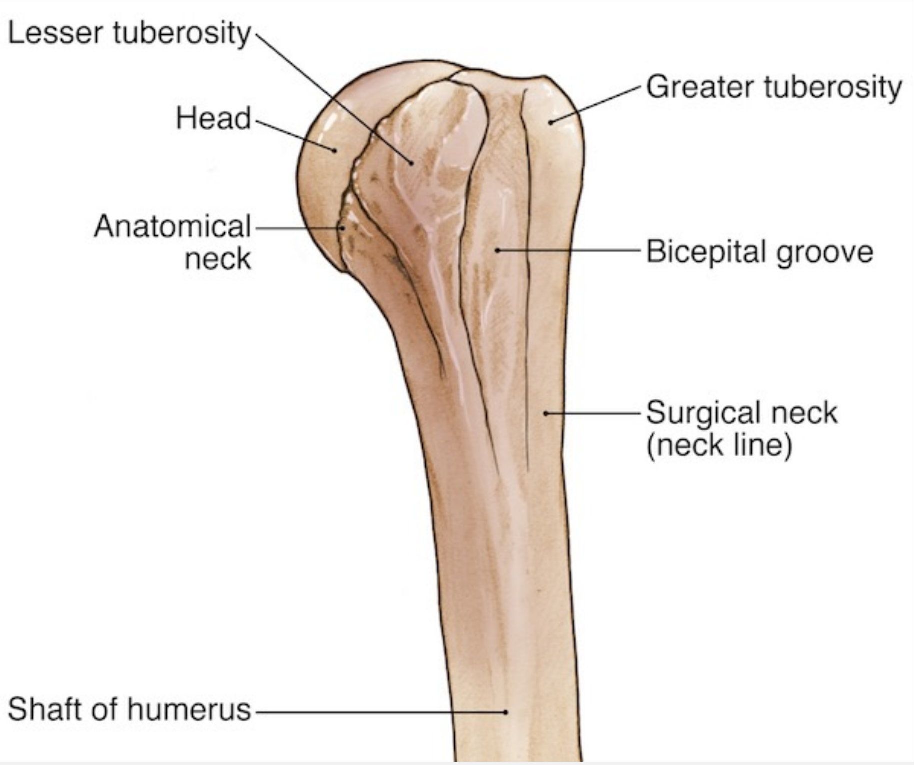


Pneumatic Bone



Bone Markings

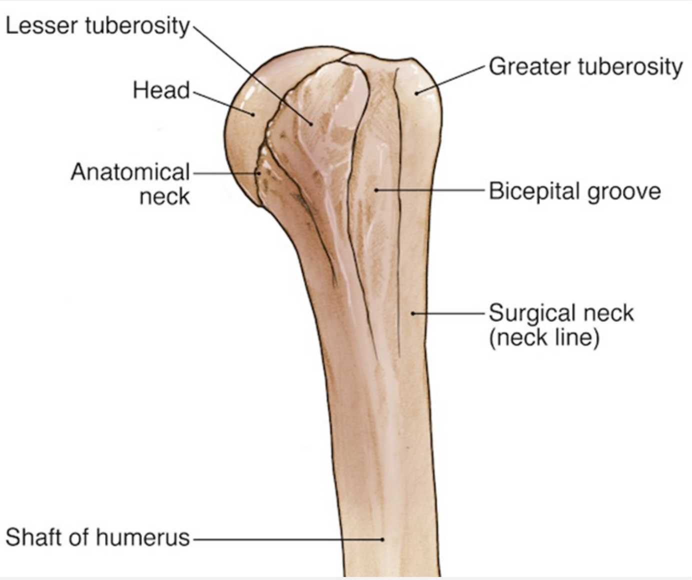
- Surface features of bones:
- Sites of attachments for muscles, tendons, and ligaments
- Passages for nerves
- and blood vessels



Bone markings

Depression & Openings:

- Fissure
- Foramen
- Fossa
- Sulcus
- Meatus



Processes:

- Condyle
- Facet
- Head
- Crest
- Epicondyle
- Line
- Spinous process
- Trochanter
- Tubercle
- Tuberocity

Skeleton system

- **Axial skeleton:** **(**80 bones)

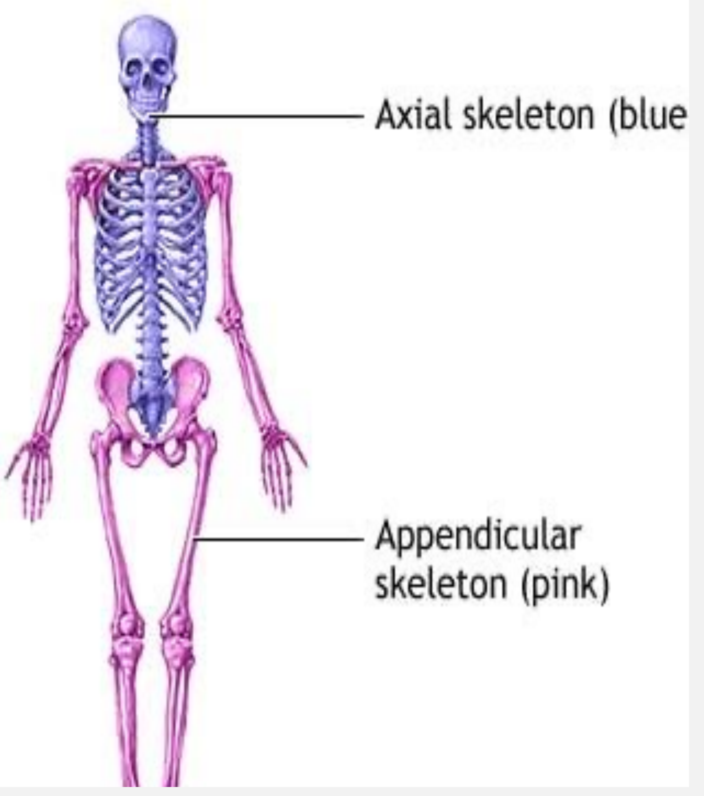
in skull, vertebrae, ribs,

sternum, hyoid bone

- **Appendicular Skeleton:**

(126 bones)- upper & lower

extremities plus two girdles



Distribution of Bones in Human Adult

- there are total of **206 bones** in an adult human.

distributed as follows:

skull (cranium + face) = 22 (8 + 14)	
Ears	**= 6**
hyoid	**= 1**
vertebral column	**= 26**
sternum	**= 1**
ribs	**= 24**
pectoral girdle and forelimbs	**= 64**
pelvic girdle and hindlimbs	**= 62**

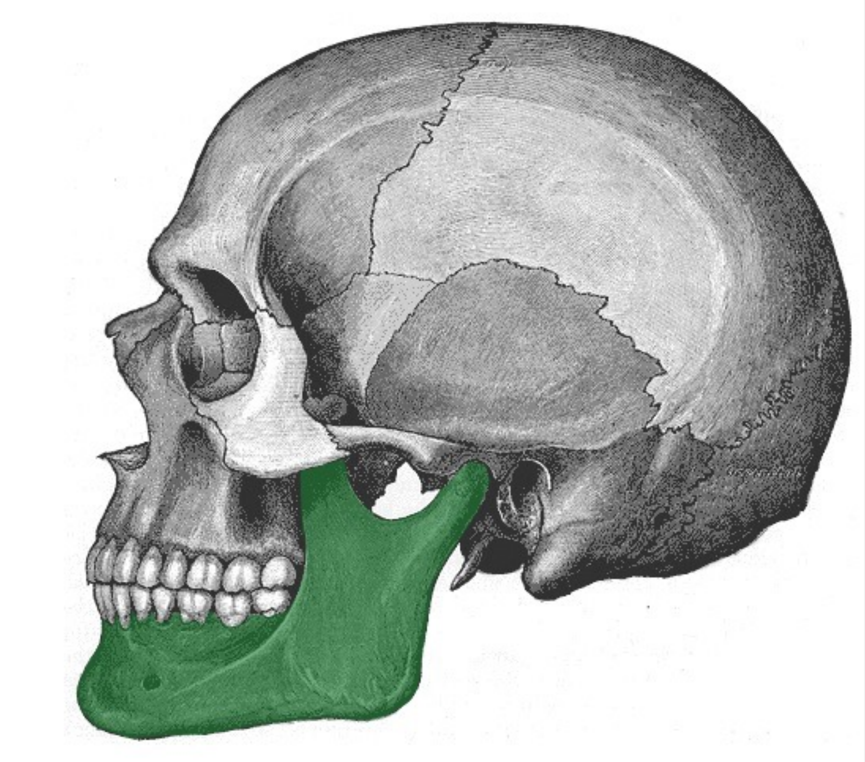
1- Skull

The skull is formed by 22 bones, and descriptively divided into:

- Upper cranium and
- lower facial skeleton.

❖ **The cranium** consists of 8 cranial bones, and contains large cranial cavity that lodges the brain and its Covering meninges.

● **The facial skeleton** is formed by 14 bones, and contains small cavities, including oral, nasal, and orbital cavities. The nasal and orbital cavities lodges organs of special sensation (smell, vision).



● The mandible is the only mobile bone of skull which articulates with temporal bones in temporo-mandibular joints (TMJ).

● All bones of the skull, except mandible, are firmly joined together by immobile, false fibrous joints called sutures.

CRINUM

Frontal

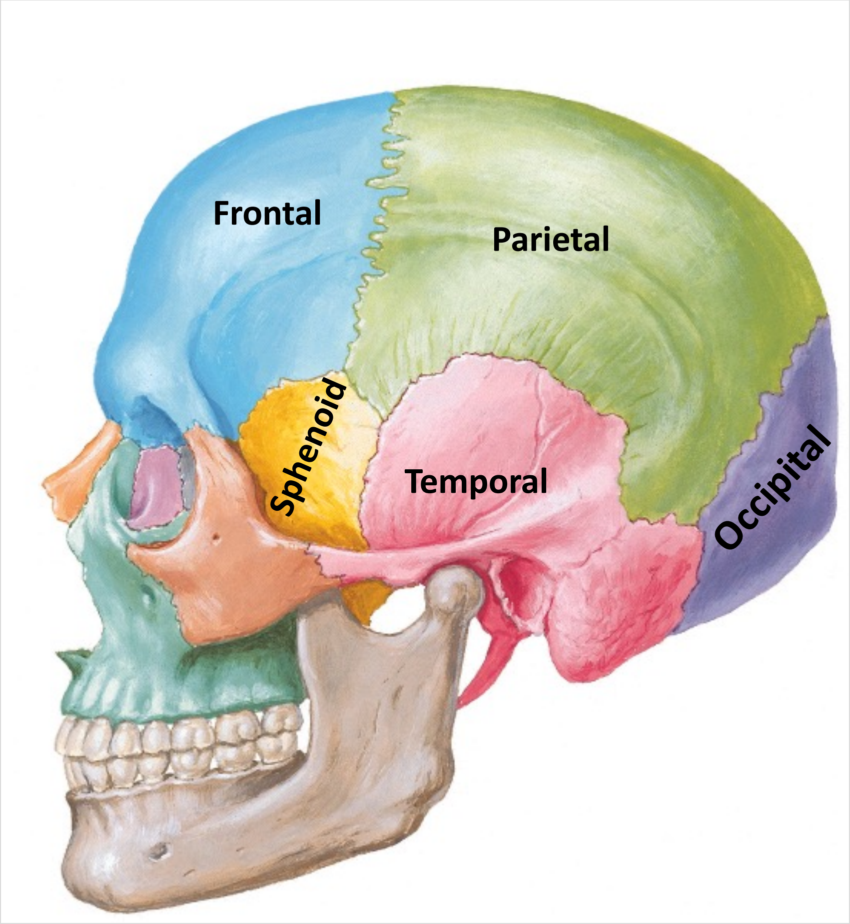
Parietal (2)

Temporal (2)

Occipital

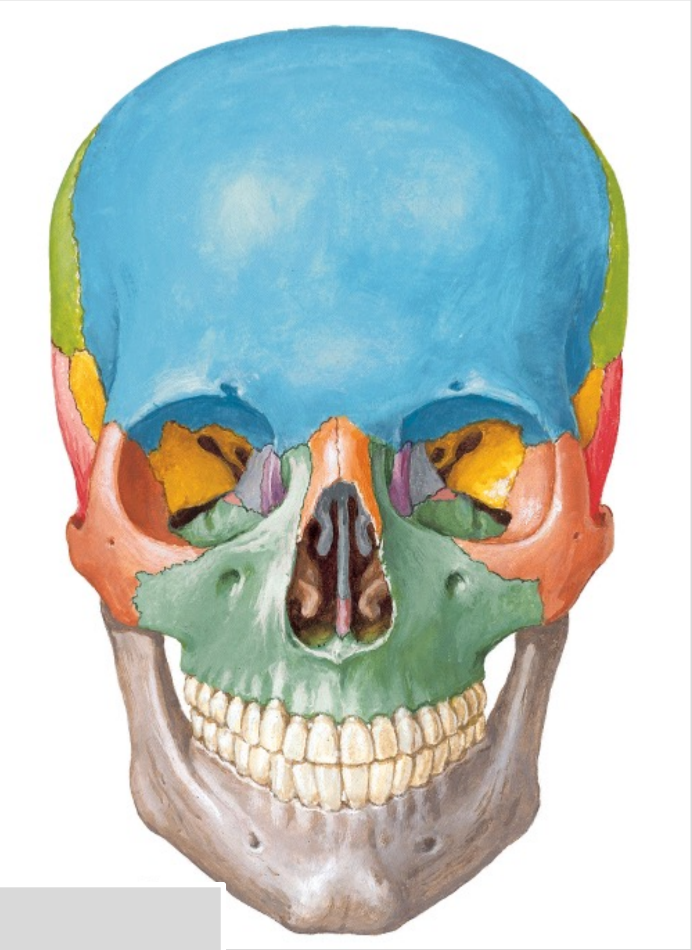
Sphenoid

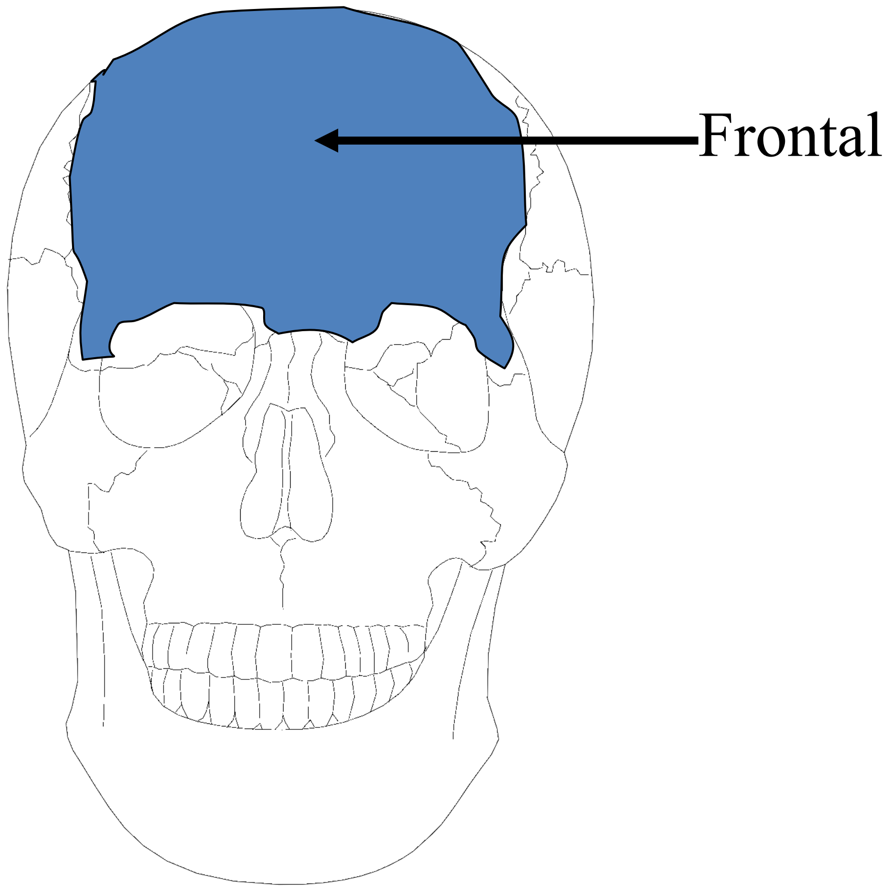
Ethmoid



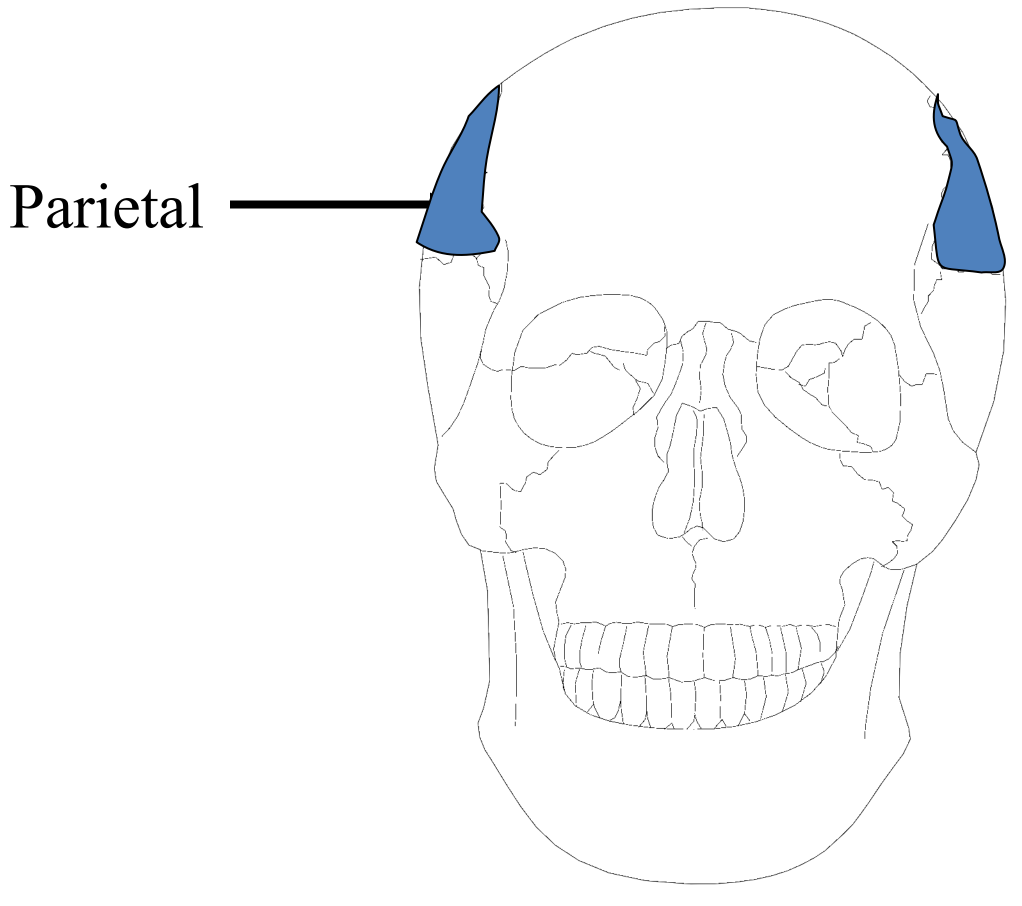
The **facial bones consist of**

- Zygomatic bones: 2
- Maxillae: 2
- Nasal bones: 2
- Lacrimal bones: 2
- Vomer: 1
- Palatine bones: 2
- Inferior conchae: 2
- Mandible: 1

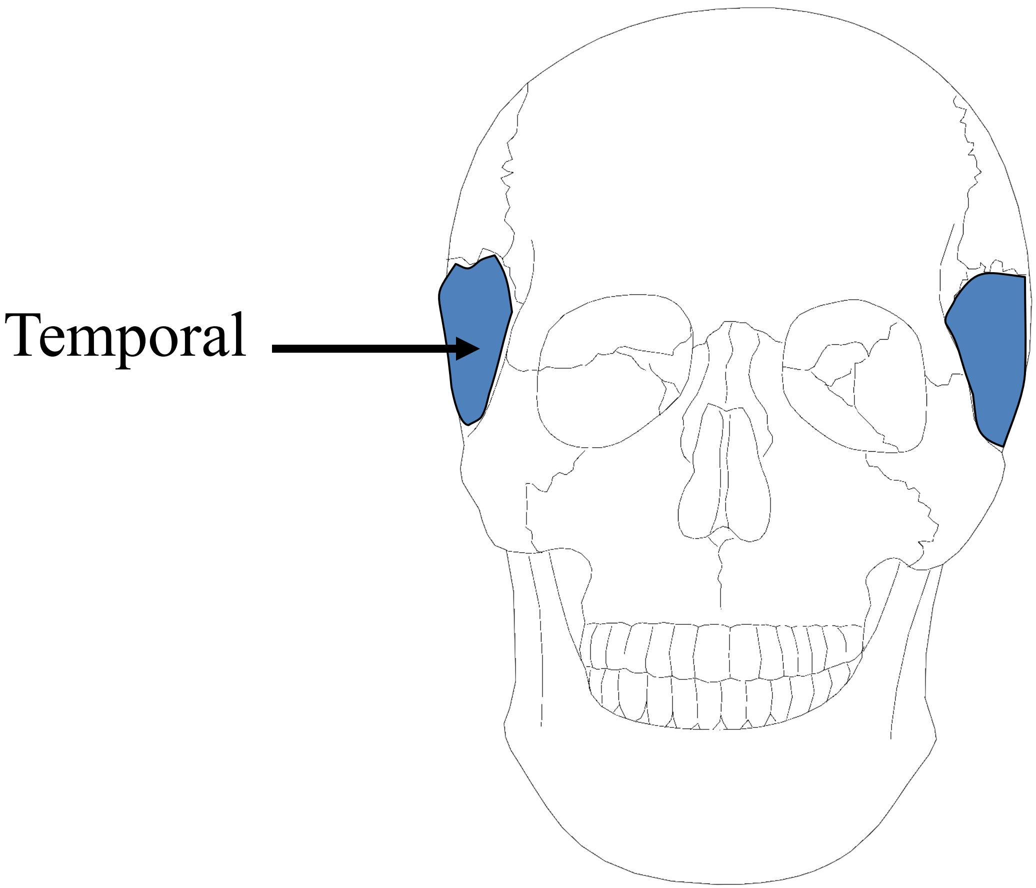




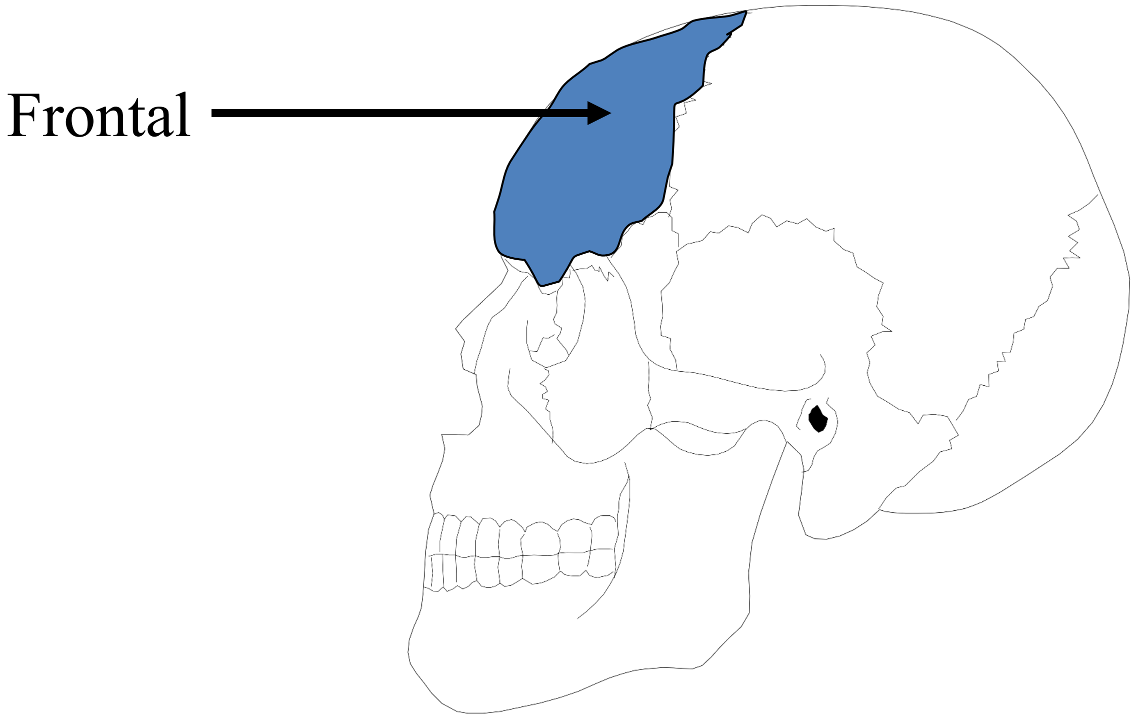
Frontal View



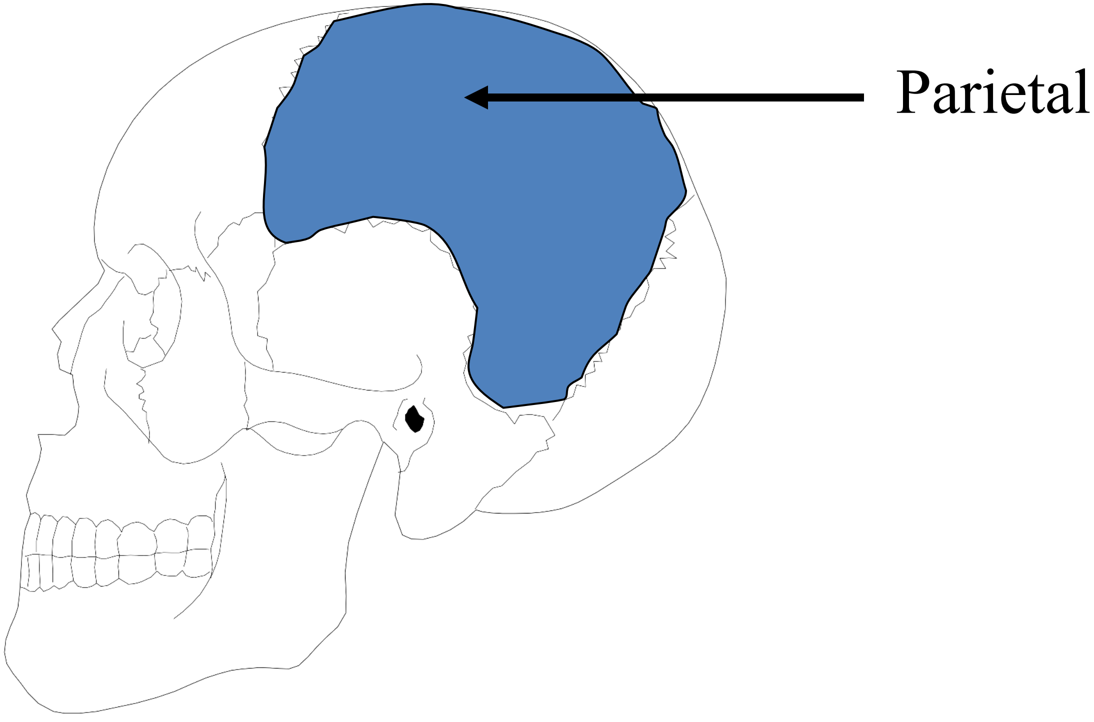
Frontal View



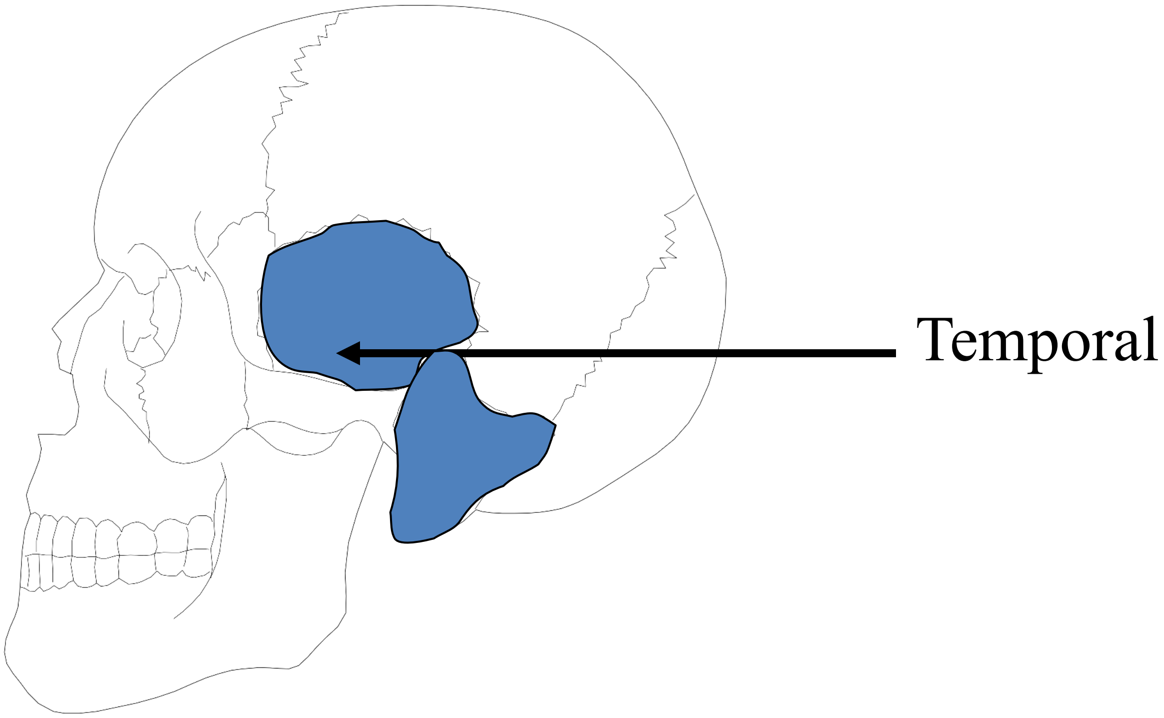
Frontal View



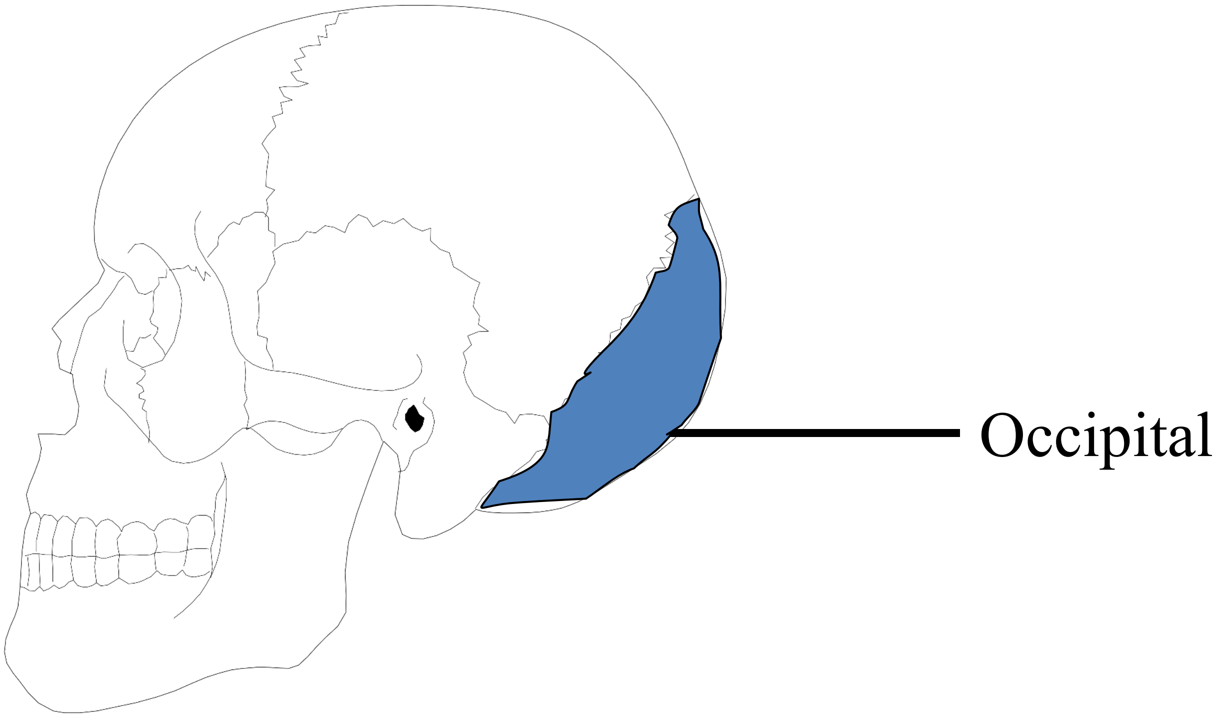
Lateral View



Lateral View



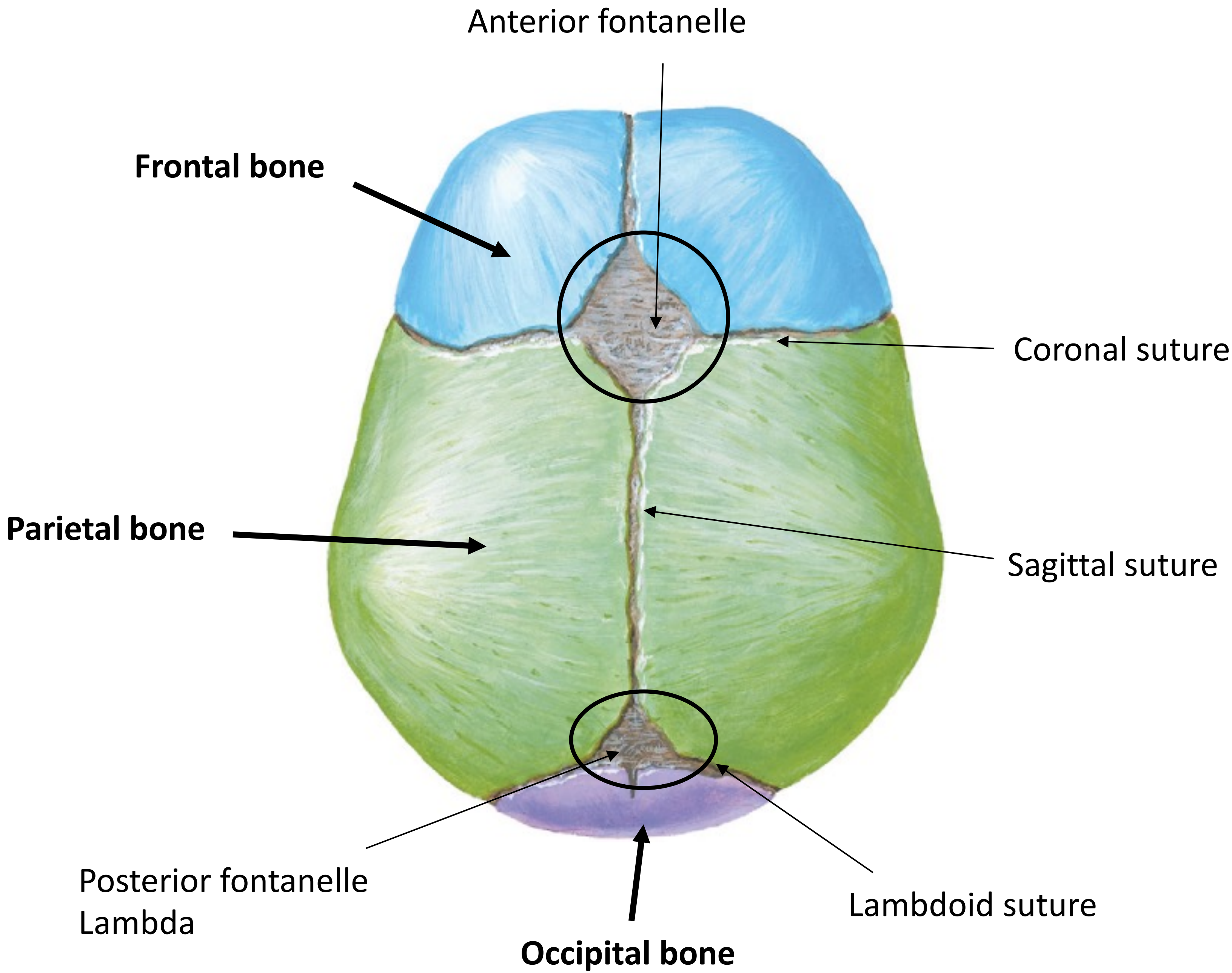
Lateral View

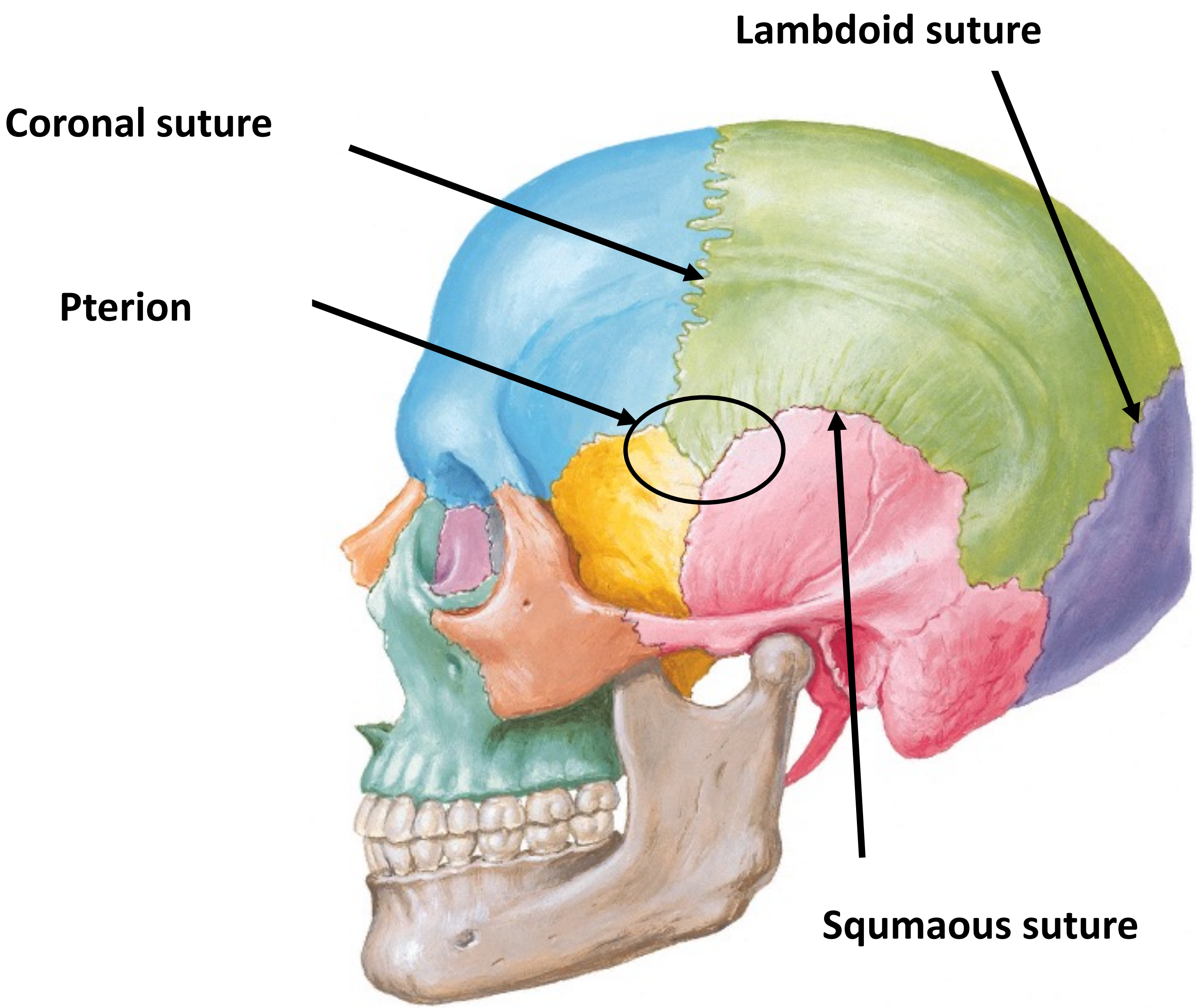


Lateral View

Cranial sutures

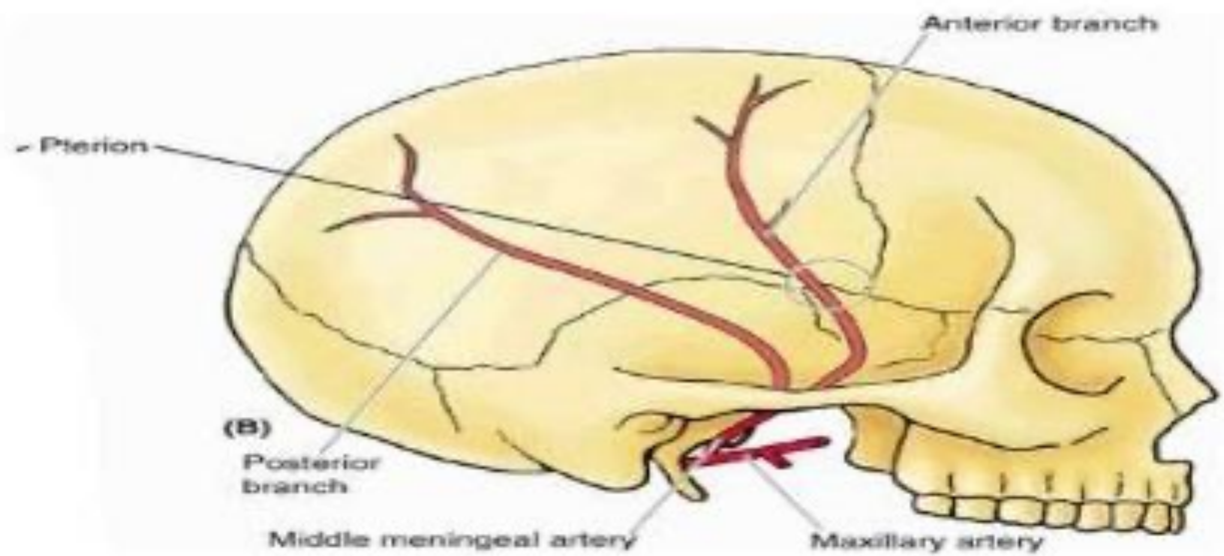
- Sagittal suture: between the parietal bones
- Coronal suture: between the frontal bone and the parietal bones
- Lambdoid suture: between the occipital bone and the parietal bones

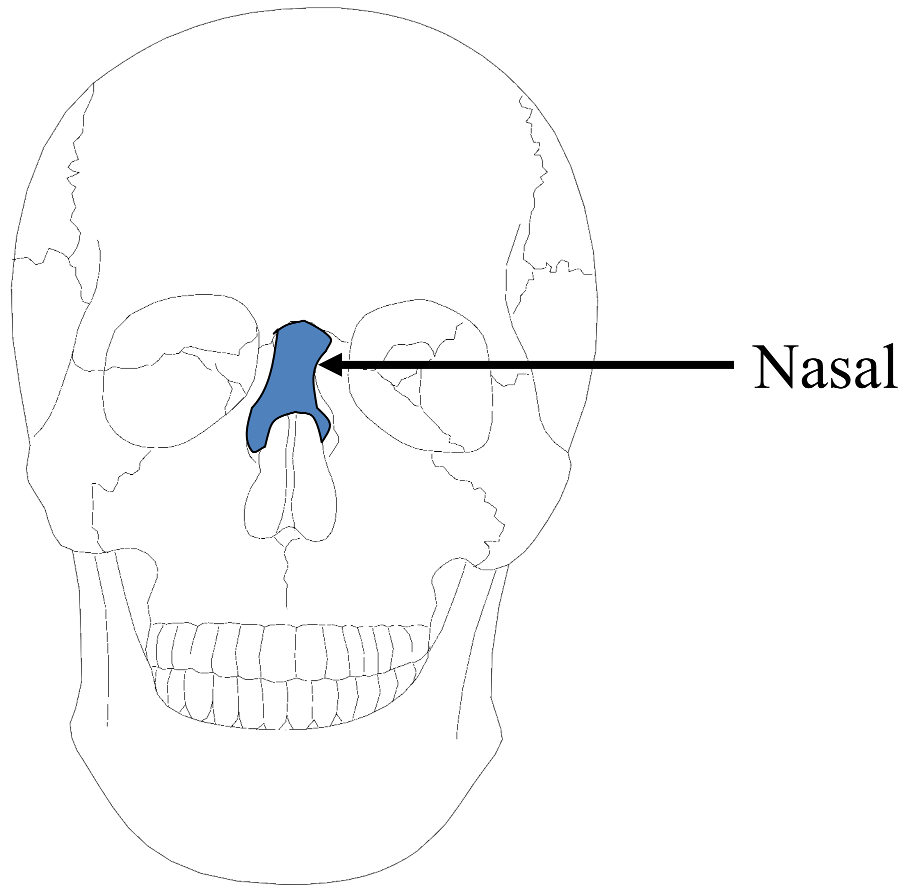

Anterior fontanelle
Frontal bone
Coronal suture
Parietal bone
Sagittal suture
Posterior fontanelle
Lambda
Lambdoid suture
Occipital bone


Lambdoid suture
Coronal suture
Pterion
Squmaous suture

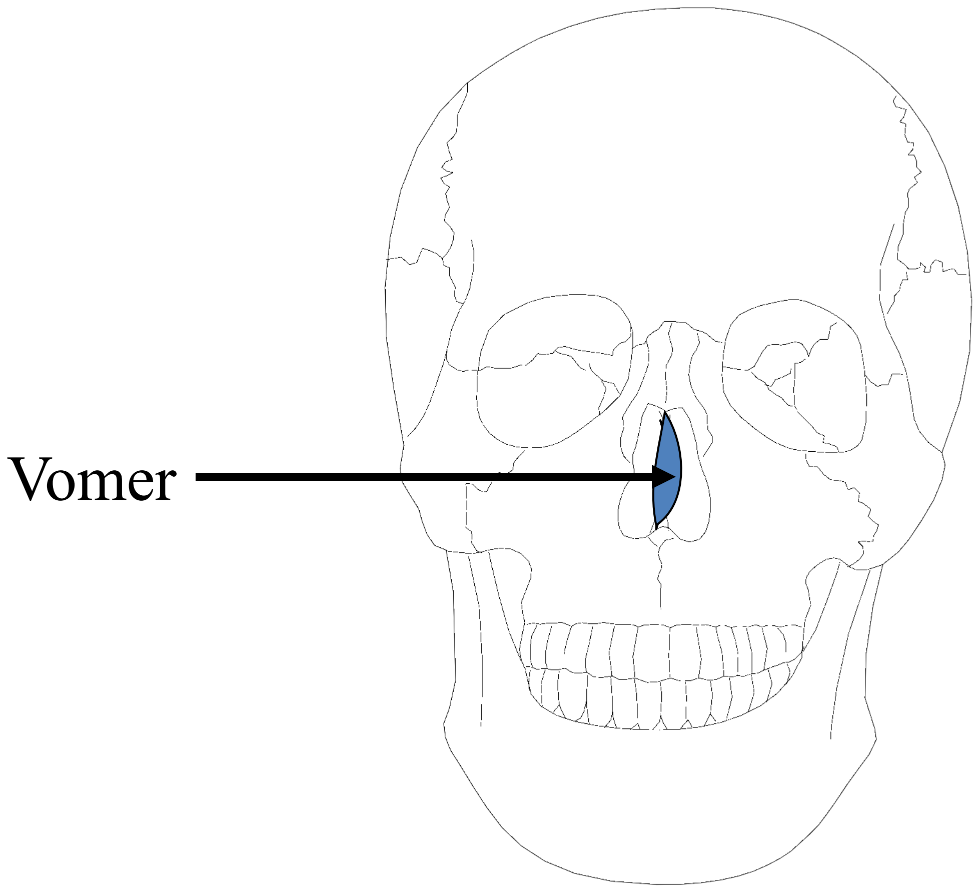
Clinical anatomy

- *The pterion is an important area because it overlies the middle meningeal artery*
- *Fracture to the pterion can rupture the anterior branch of this artery resulting in hematoma which exerts pressure on the underlying cerebral cortex*
- *An untreated meningeal artery can cause hemorrhage which can lead to death in a few hours*

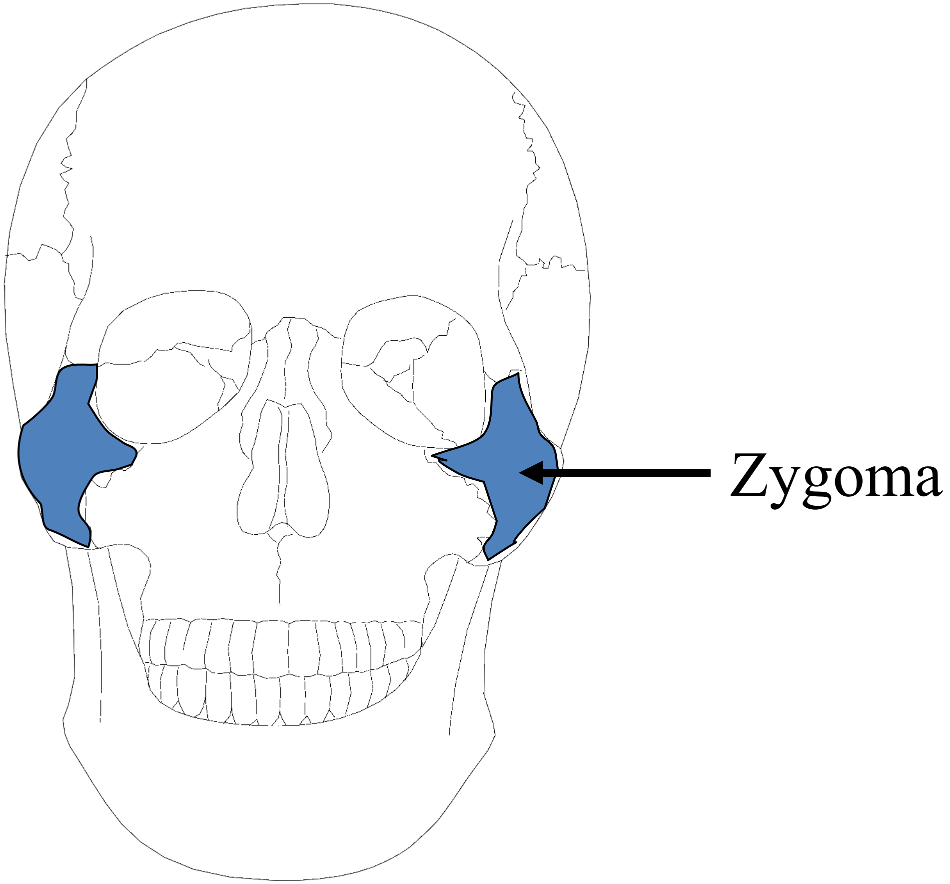




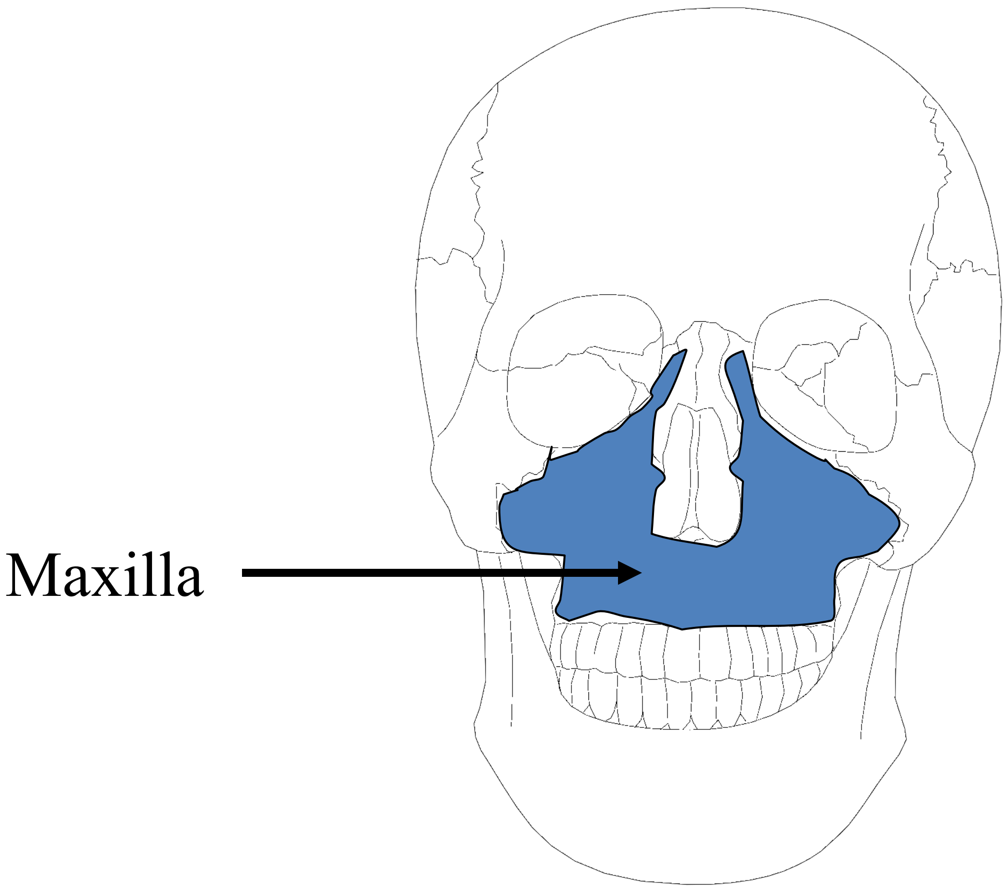
Frontal View



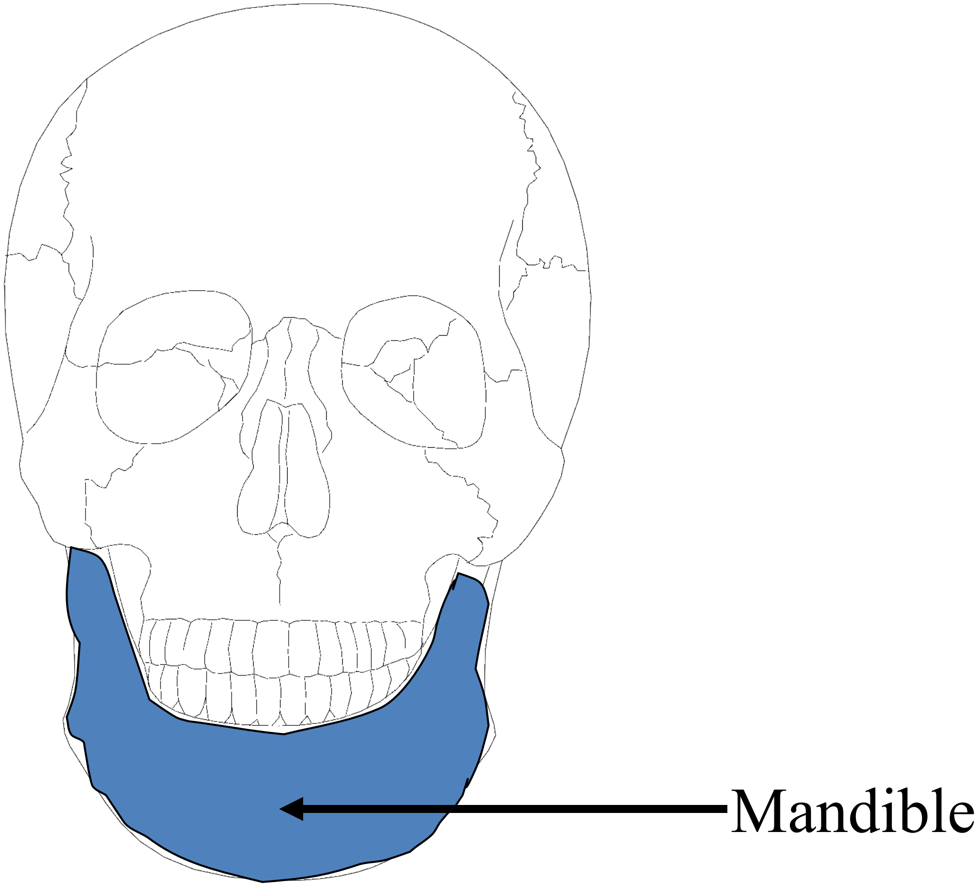
Frontal View



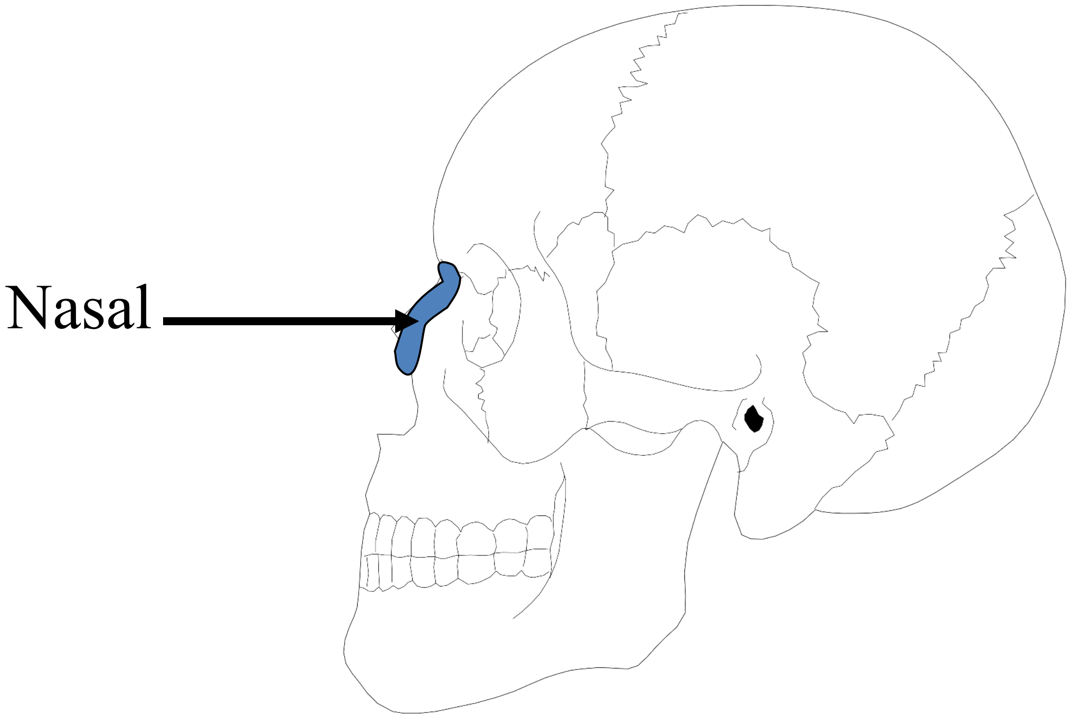
Frontal View



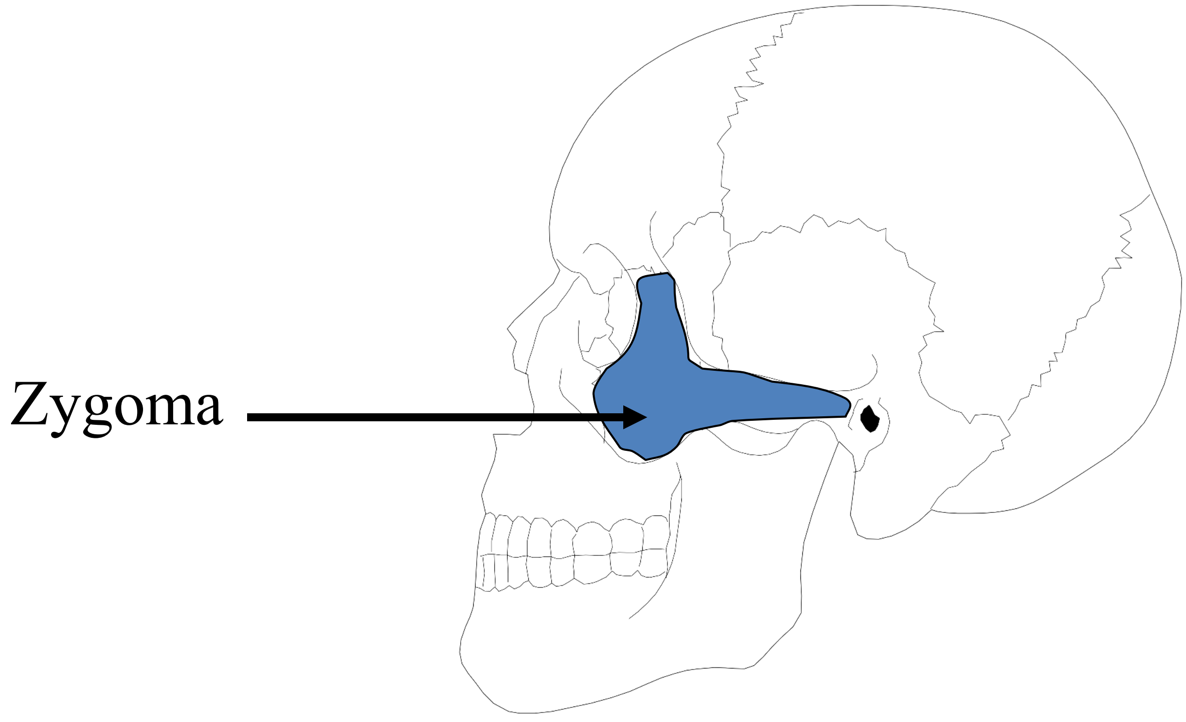
Frontal View



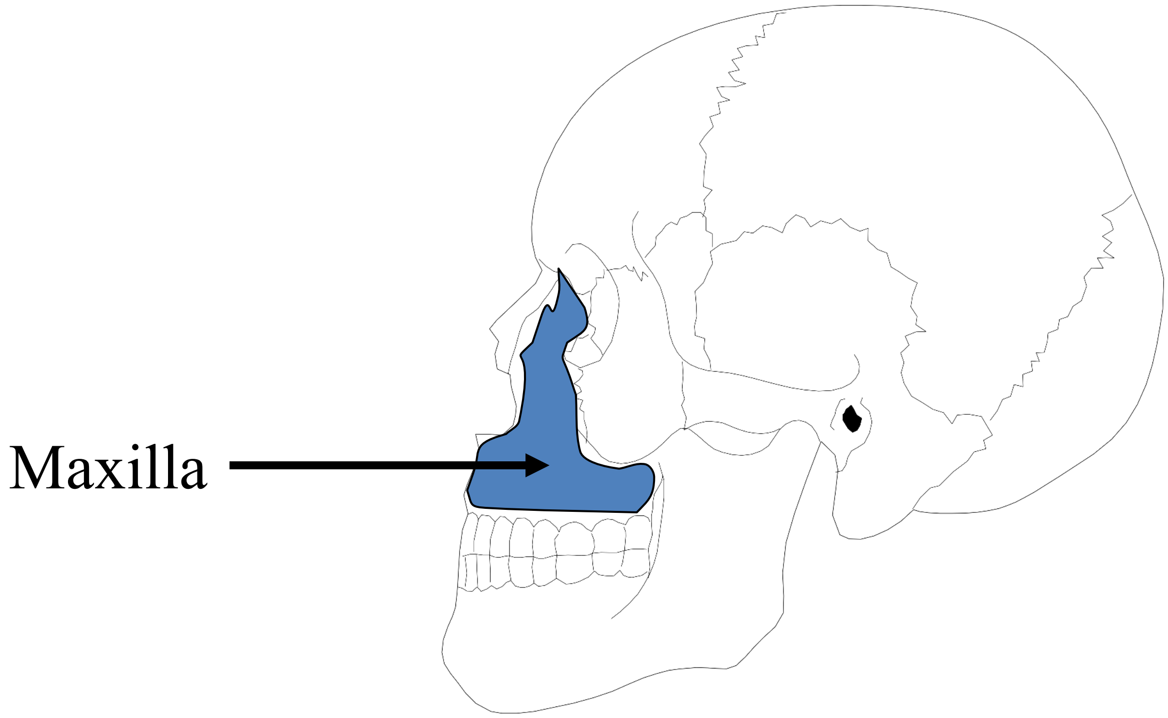
Frontal View



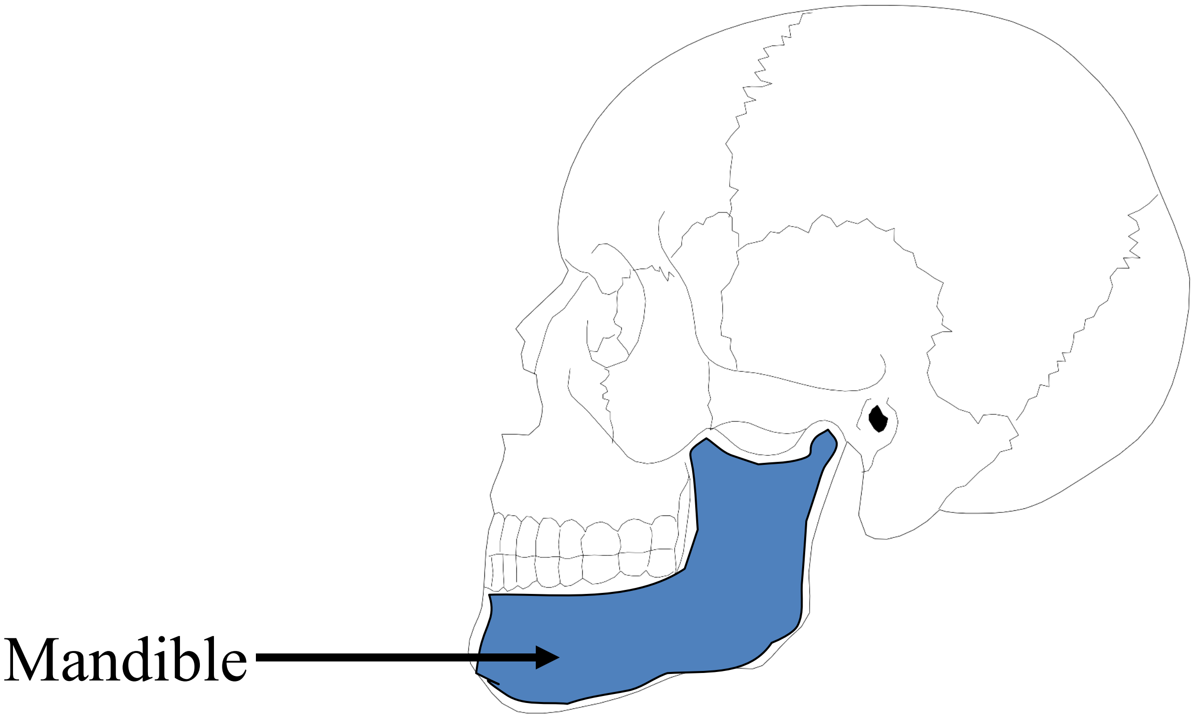
Lateral View



Lateral View



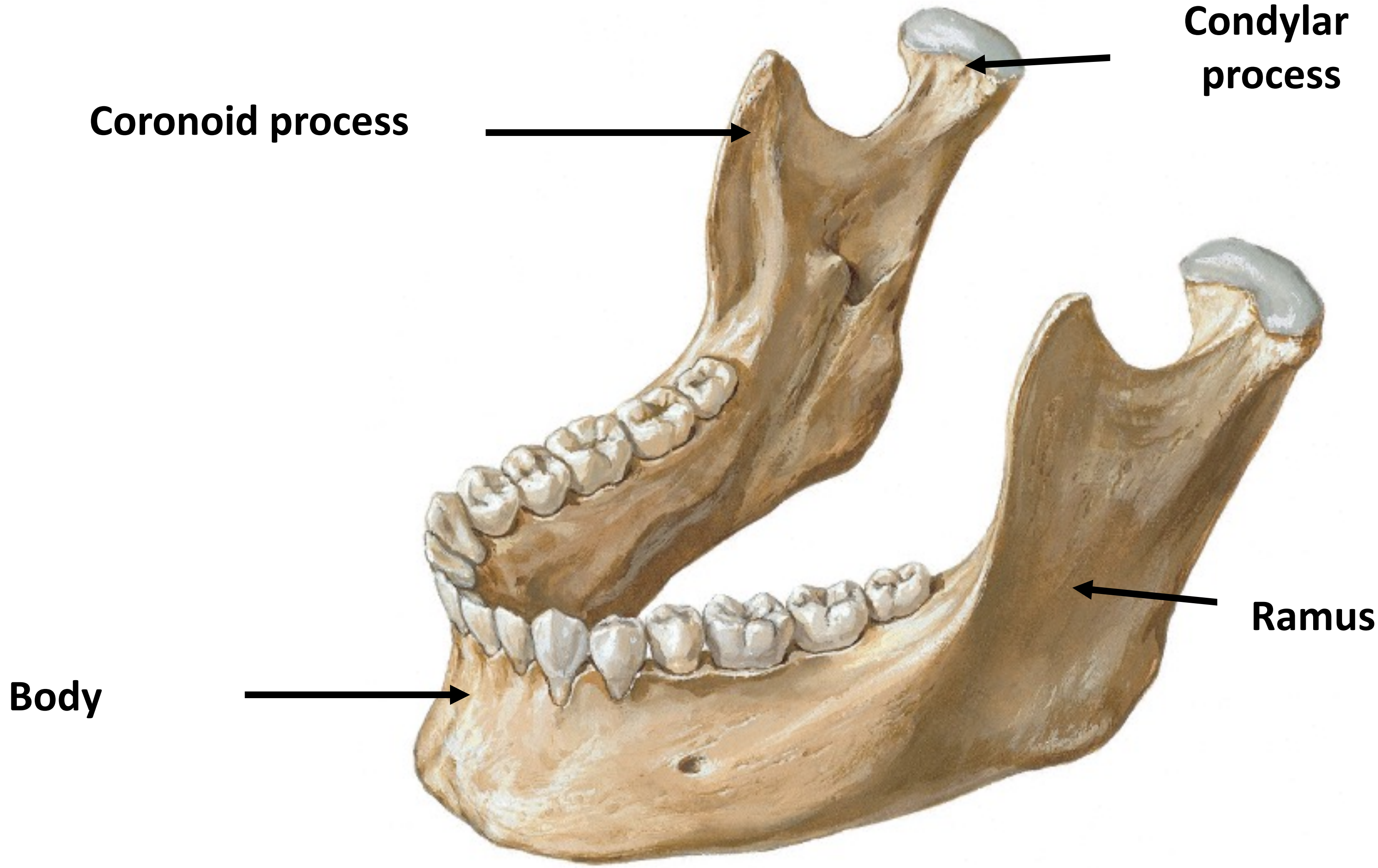
Lateral View



Lateral View

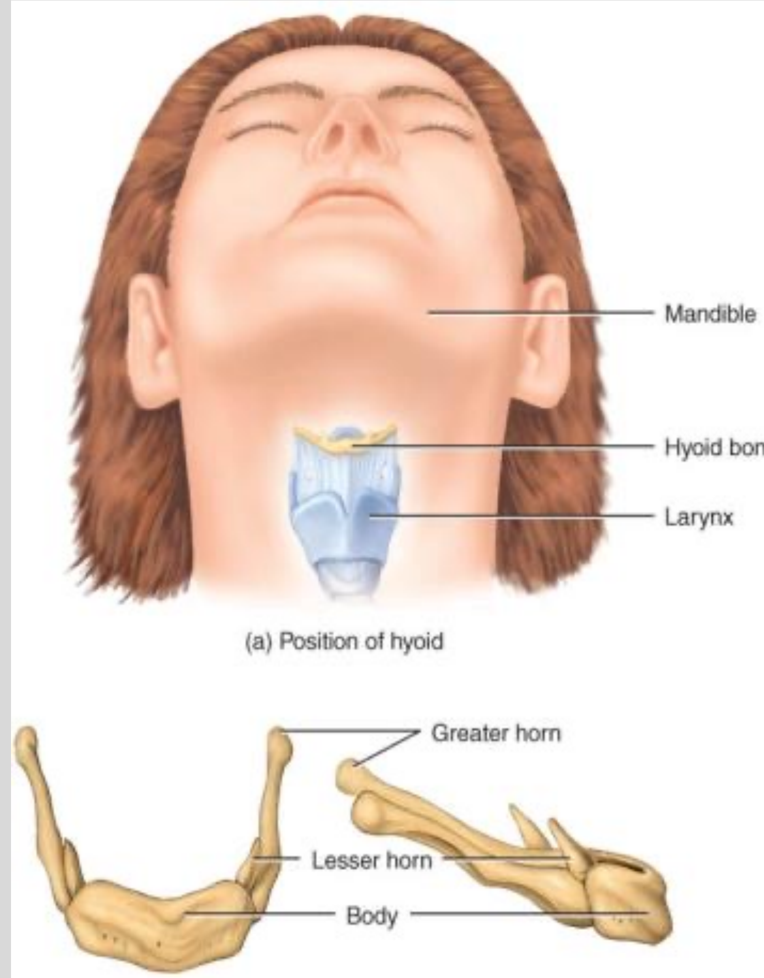
Mandible

Anterolateral Superior View



Hyoid Bone

- U-shape located in the neck.
- Has body, greater and lesser horns.
- Does not articulate with any bone and suspended in place by membranes, muscles and ligaments.
- Important for movements of tongue, pharynx and larynx .



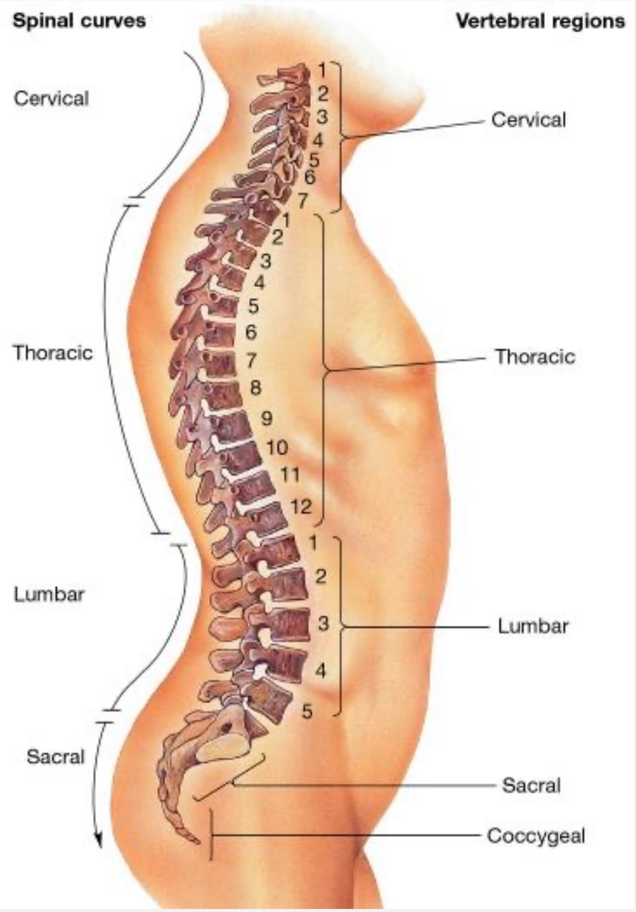
(a) Position of hyoid
(b) Anterior view
(c) Right lateral view

The vertebral column

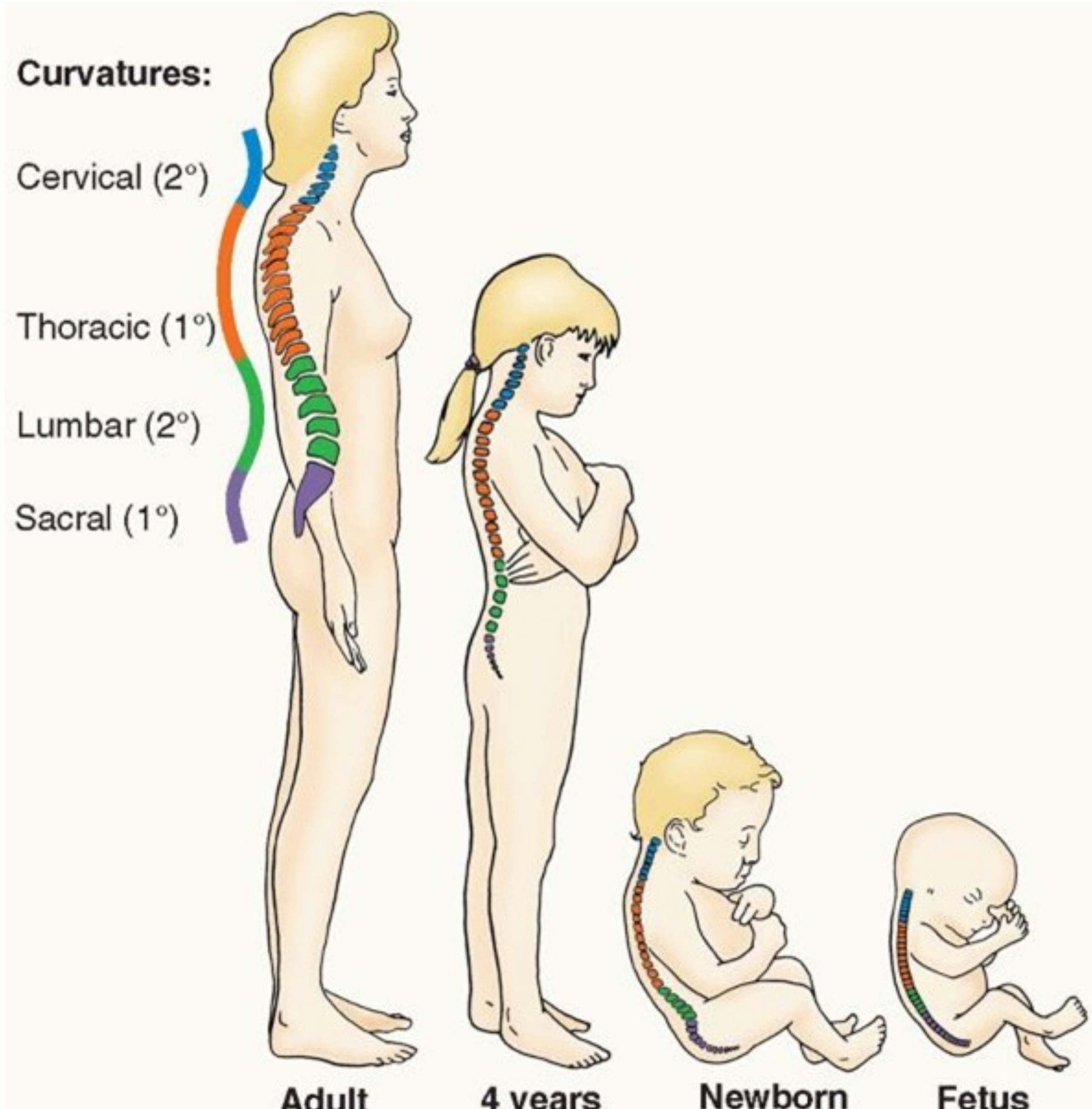
- Its also called spinal column
- It supports the body weight and transmits the weight to the ground
- It S shaped in adult that extends from skull to tip of coccyx
- Formed by large no of bone which are irregular in shape called vertebrae

The vertebral column

- There are 33 vertebrae divided into five groups
- *Seven cervical*
- *Twelve thoracic*
- *Five lumber*
- *Five fused sacrum*
- *Four fused coccygeal*



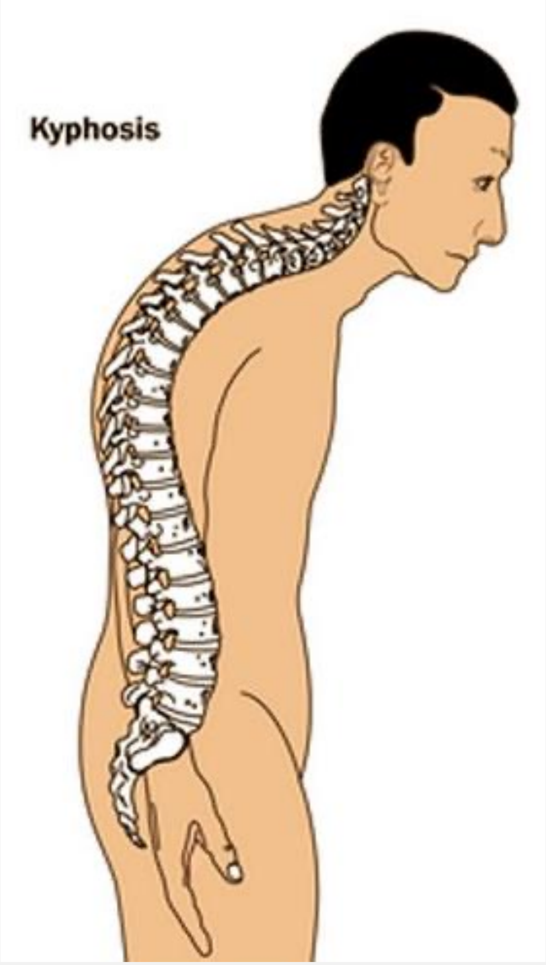
Curves of vertebral colunm



Abnormal curvature

- **Kyphosis (hump back)**

increase in the thoracic
curvature

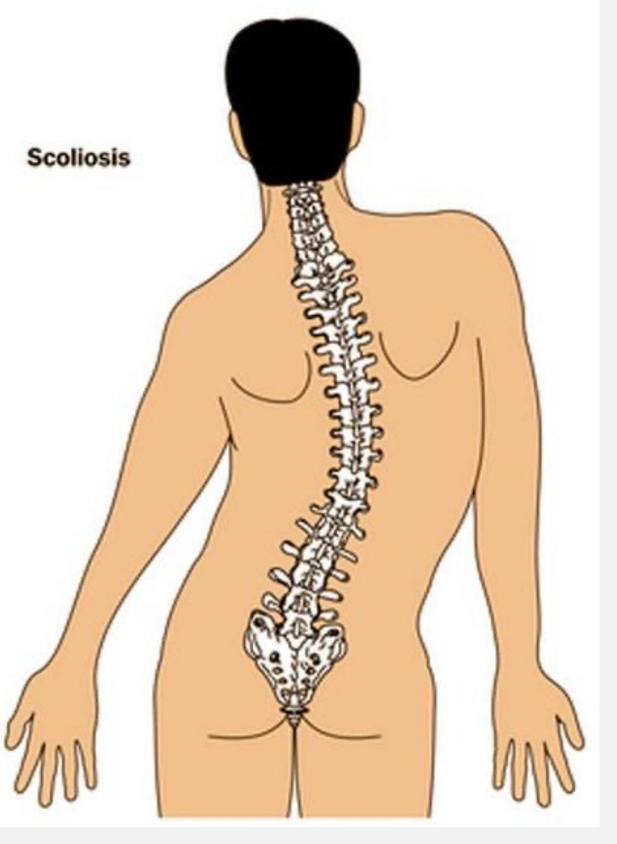


lordosis

- Increase in the lumber curvature

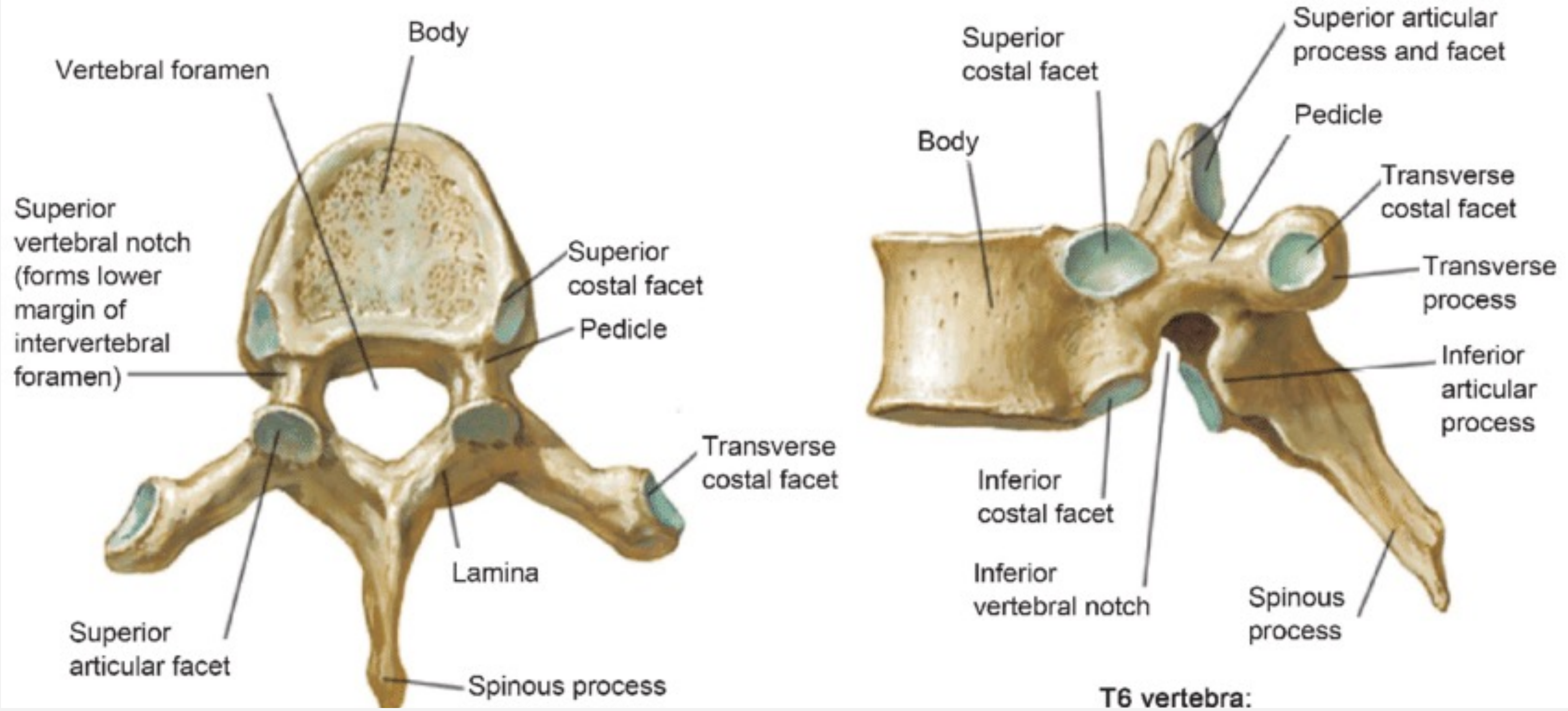
Scoliosis

- Abnormal lateral curvature of the vertebral column



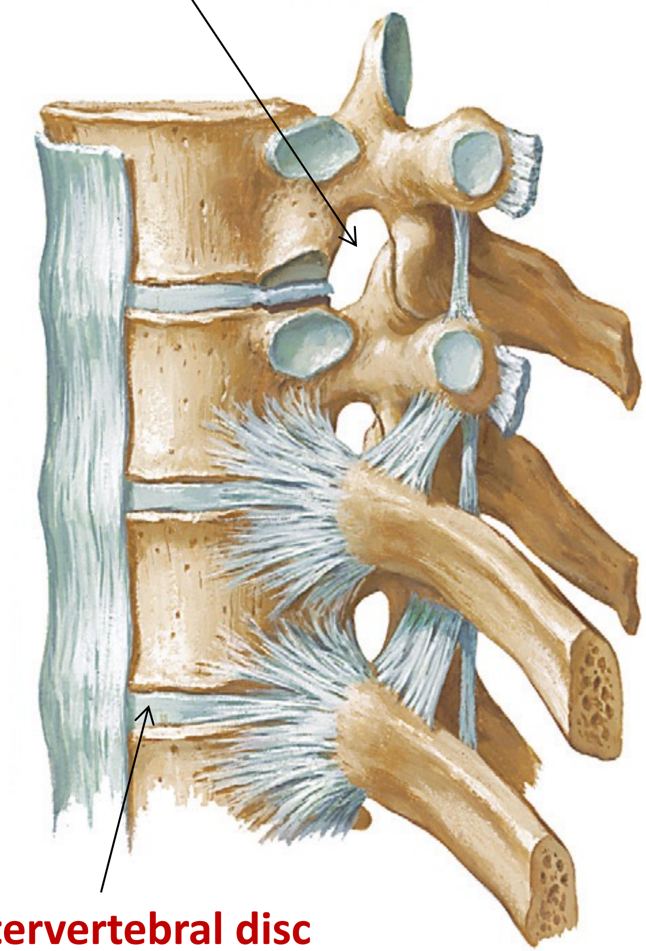
Typical Vertebrae

- Body
- Vertebral arch   pedicle   laminae
- Vertebral foramen
- Seven processes. 2 transverse , 1 spinous 4 articular



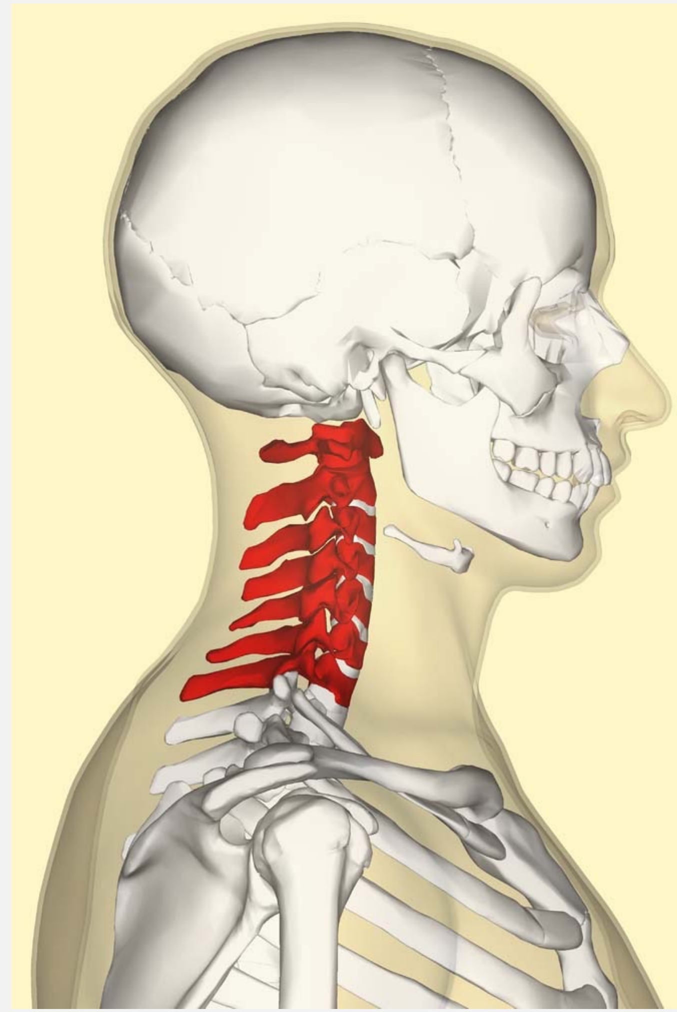
Intervertebral Foramen

- Neural foramen
- Notches between adjacent vertebrae.
- Allows for the passage of the spinal nerve roots, spinal artery, veins, nerve plexus, and ligaments



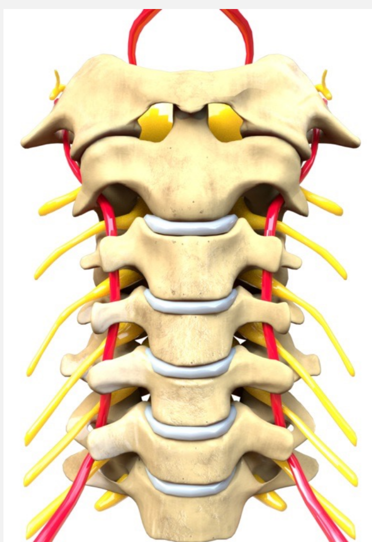
Cervical vertebrae

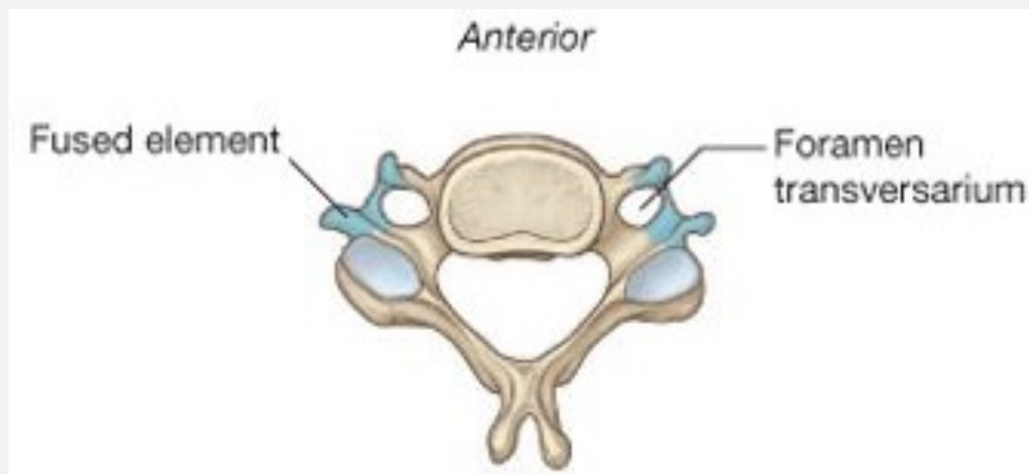
- There are seven cervical vertebrae
- The 3rd to 6th are typical
- The 1st to 2nd are modified to permit movement of the head
- The 7th shows some features of the thoracic vertebrae



Characteristics of a Typical Cervical Vertebra

- The transverse processes possess a **foramen transversarium** for the passage of the vertebral artery and veins ***(note that the vertebral artery passes through the transverse processes C1 to 6 and not through C7).***
- The spines are small and bifid.
- The body is small and broad from side to side.
- The vertebral foramen is large and triangular.





Cervical Vertebrae

- **Atlas – 1st; supports head has no body and spine**
- **Axis – 2nd; dens pivots to turn head,** Identified by the presence of strong tooth like process called Dens

Atlas – 1st

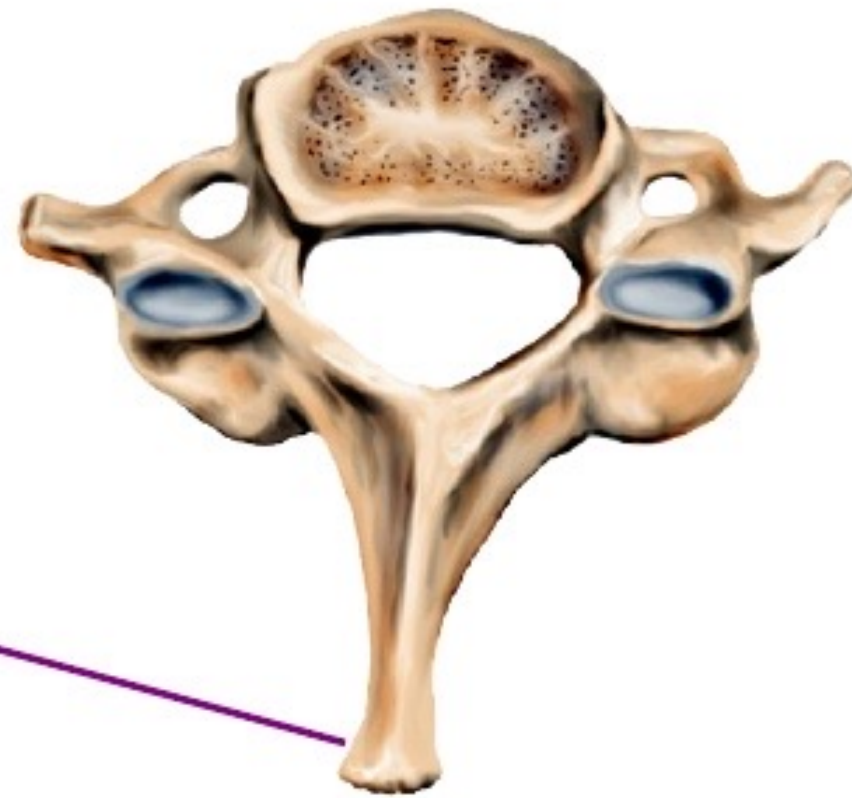
Axis – 2nd; dens pivots

Vertebra Prominens (C7)

C7 is referred to as the VERTEBRA PROMINENS because it has a longer and larger spinous process than the other cervical vertebrae.

This spinous process is not usually bifid.

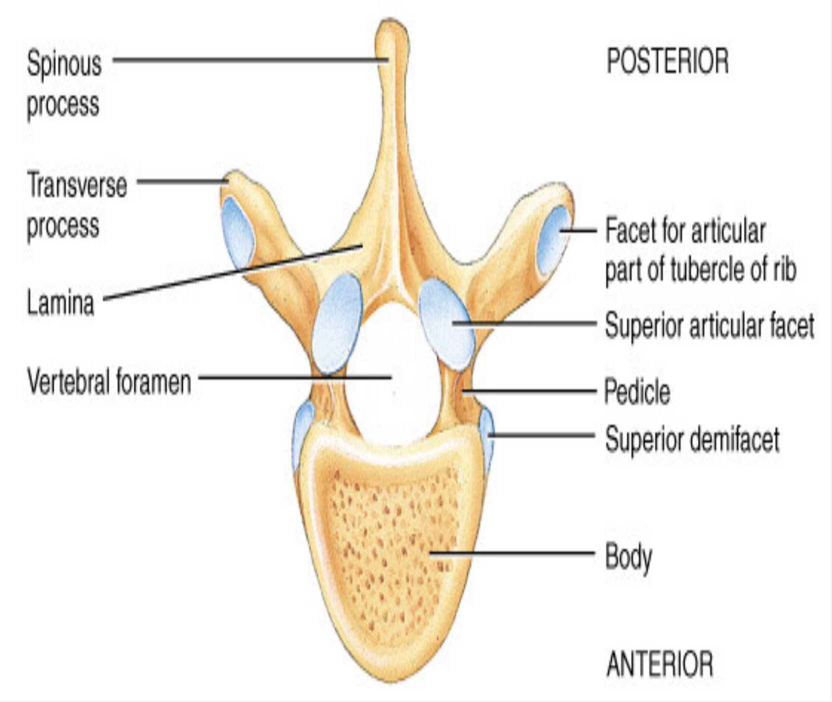
Narrow intervertebral foramina
Nerve roots at risk of compression



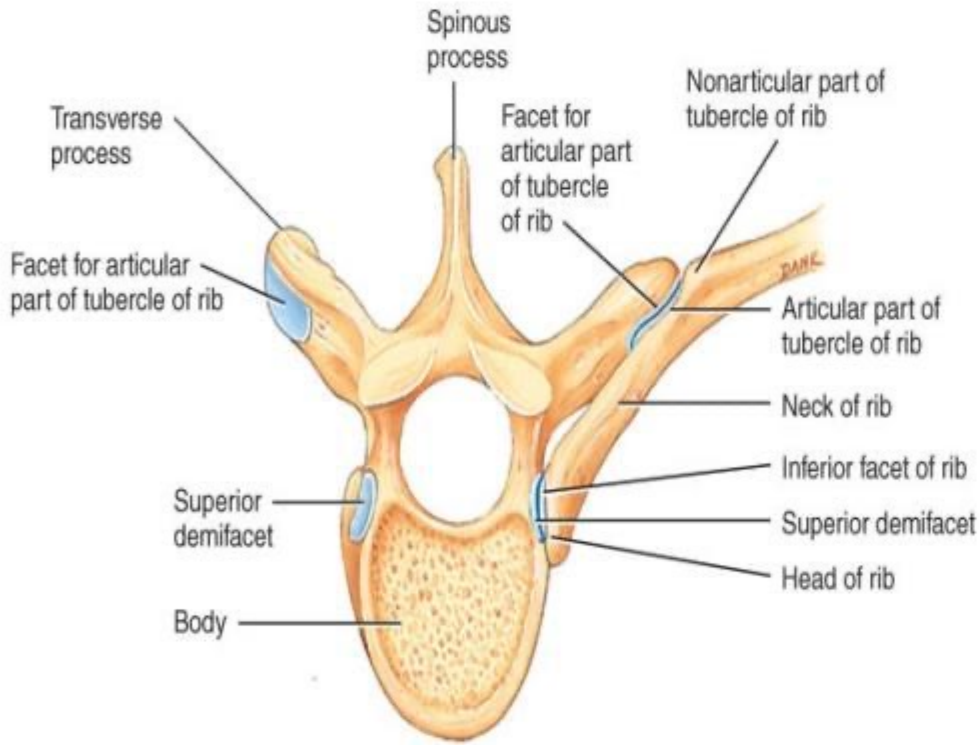
Axial View

Thoracic Vertebrae (T2-T8)

- All articulate with ribs
- Have heart-shaped bodies
- Each side of the body bears demifacets for articulation with ribs



Rib Articulation



Each typical rib articulates with vertebral bodies and tubercle of transverse process

Lumber vertibrae

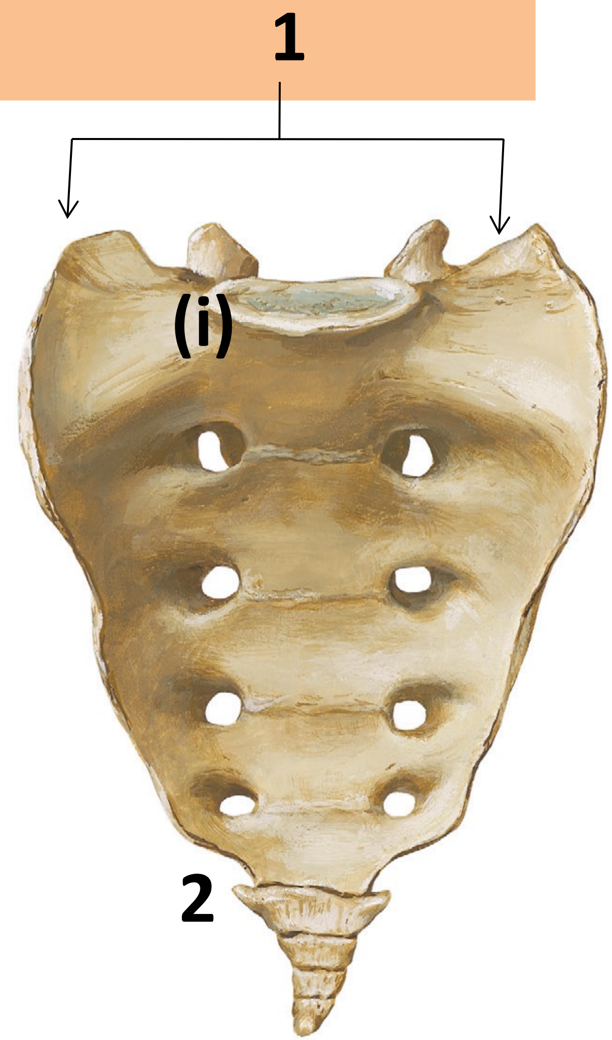
- There are five lumber vert

1. The body is large
2. Transverse processes are long and slender
3. The vertebral arch are thick, short and strong
4. The spine processes are short
5. The vertebral foramen are triangle

sacrum

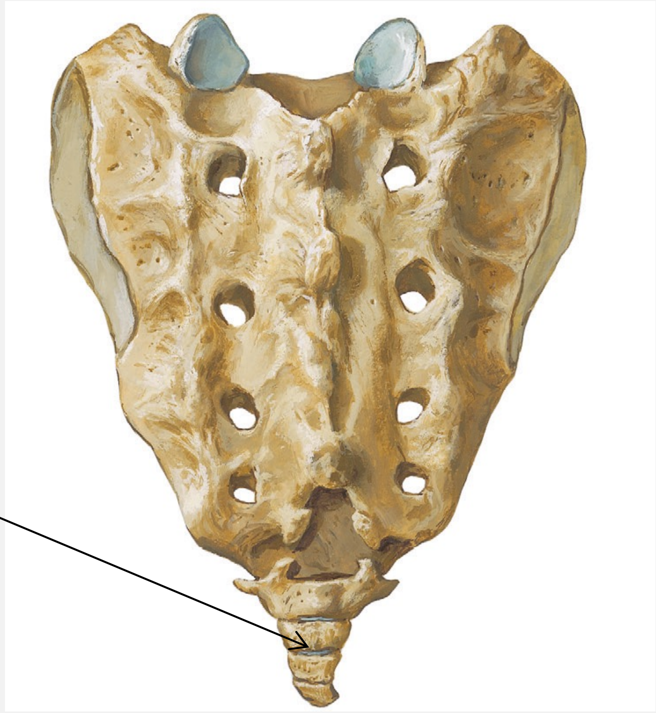
- The sacrum is a large triangular bone formed by five rudimentary vertebrae It has

1. **Base** which formed by the 1^{st} sacral vertebra and has ***(i)*** ***sacral promontry***
2. **Apex** formed by 5^{th} sacral vert
3. **Pelvic and dorsal surfaces**



coccyx

- Is a small triangular formed by fused of four rudimentary coccygeal vertebrae

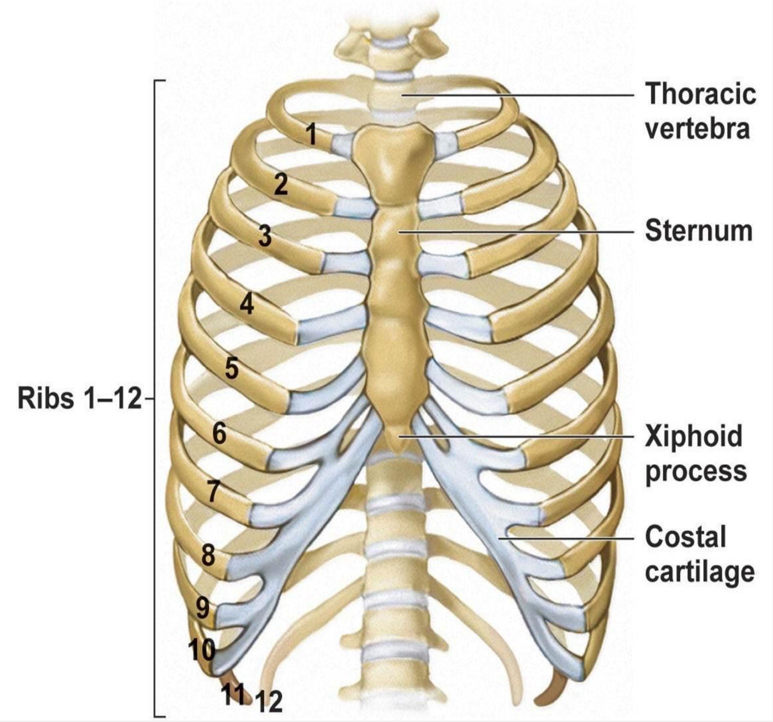


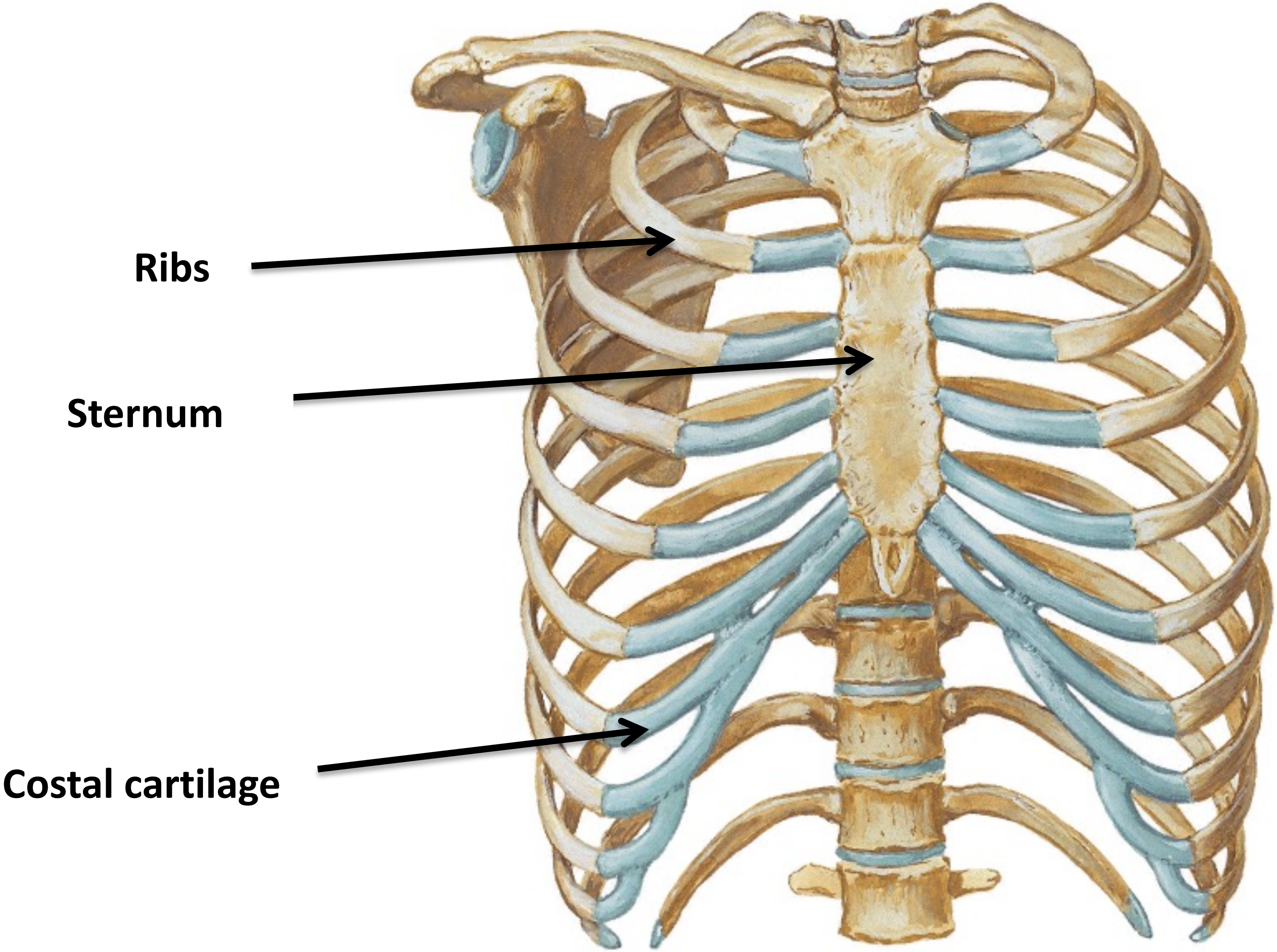
Skeleton of the thorax

- Thoracic cage is an osteo-cartilagenous conical cage which has a narrow inlet & a wide outlet **?**

Boundaries of thoracic cage

- **Ant:** Sternum, Costal cartilages and ribs.
- **Post:** Thoracic vertebrae and ribs.
- **Lat:** Ribs.

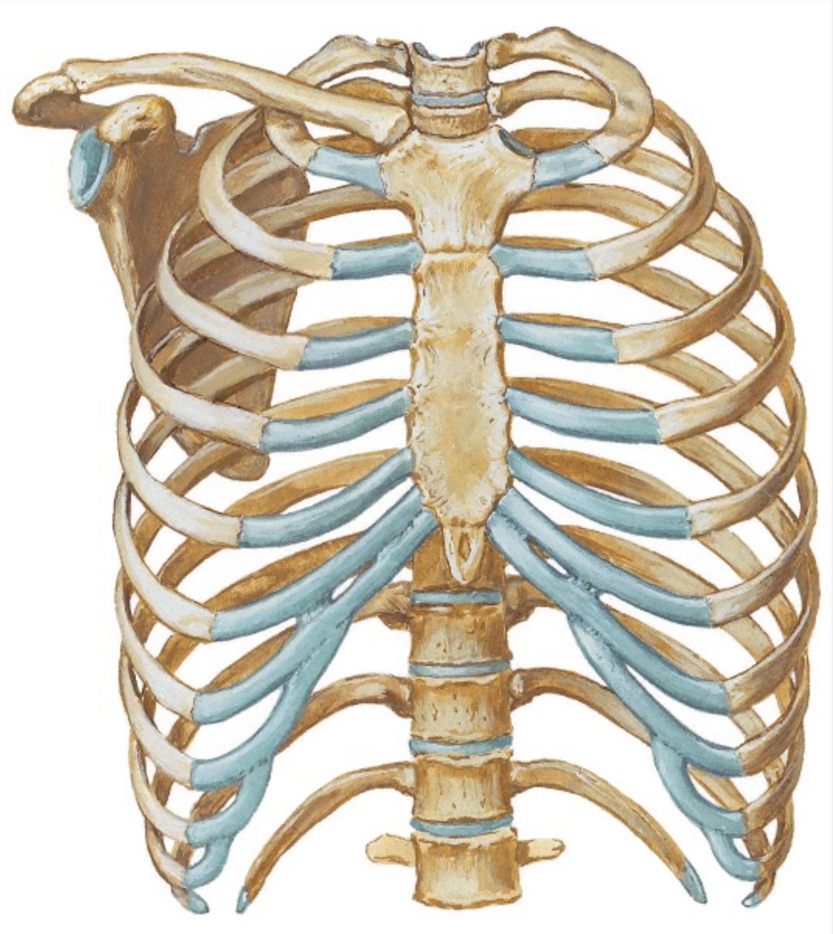



Ribs
Sternum
Costal cartilage

Ribs

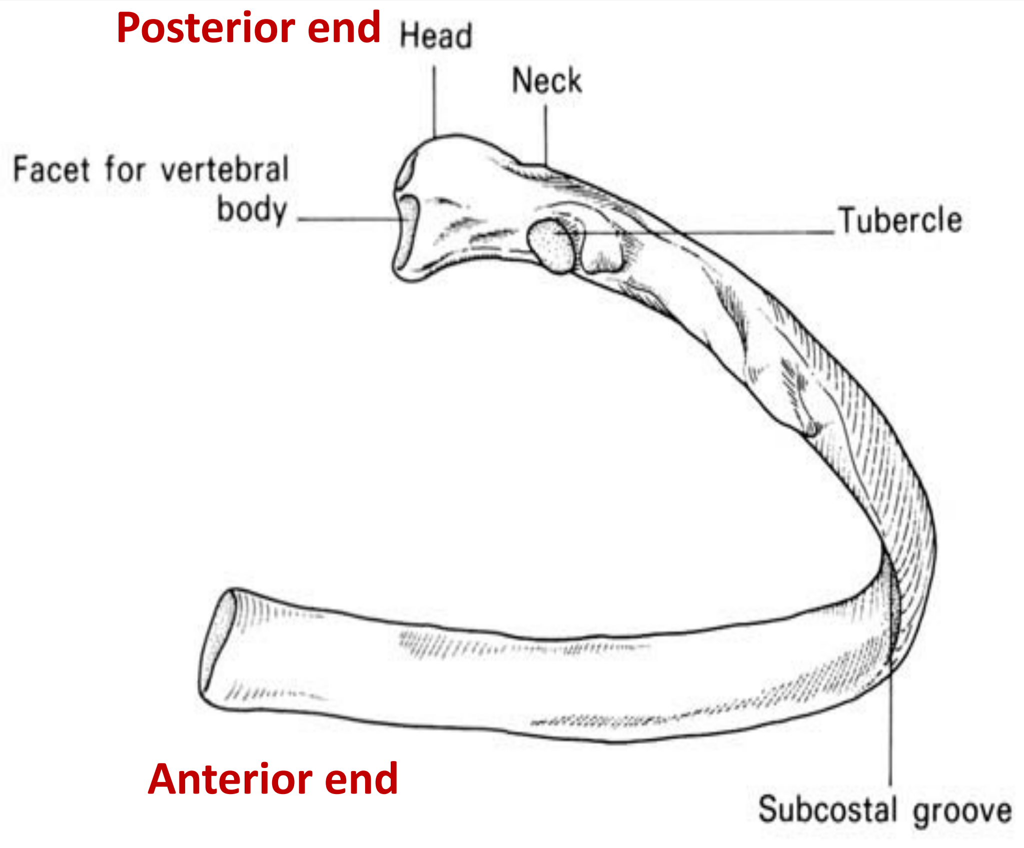
- 12 pairs of ribs

- Ribs 1-7 are true ribs
- Ribs 8-10 are false ribs
- Ribs 11-12 are floating



Typical rib

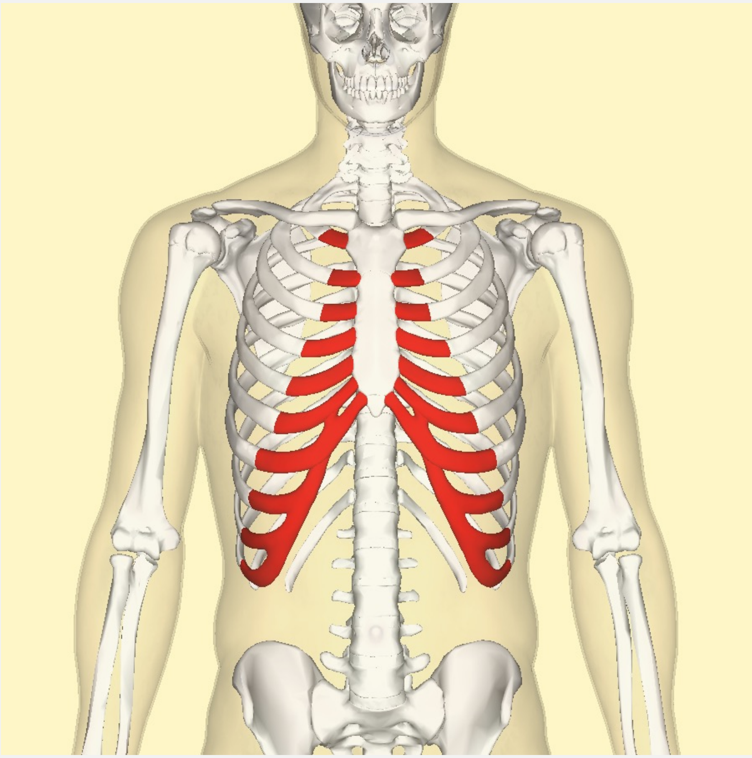
1. Head
2. Neck
3. Tubercle



The costal cartilage

- Are un ossified ant part of the ribs
- They are made up of hyaline cartilage
- From the 7^{th} to 10^{th}

forming costal margin



Sternum

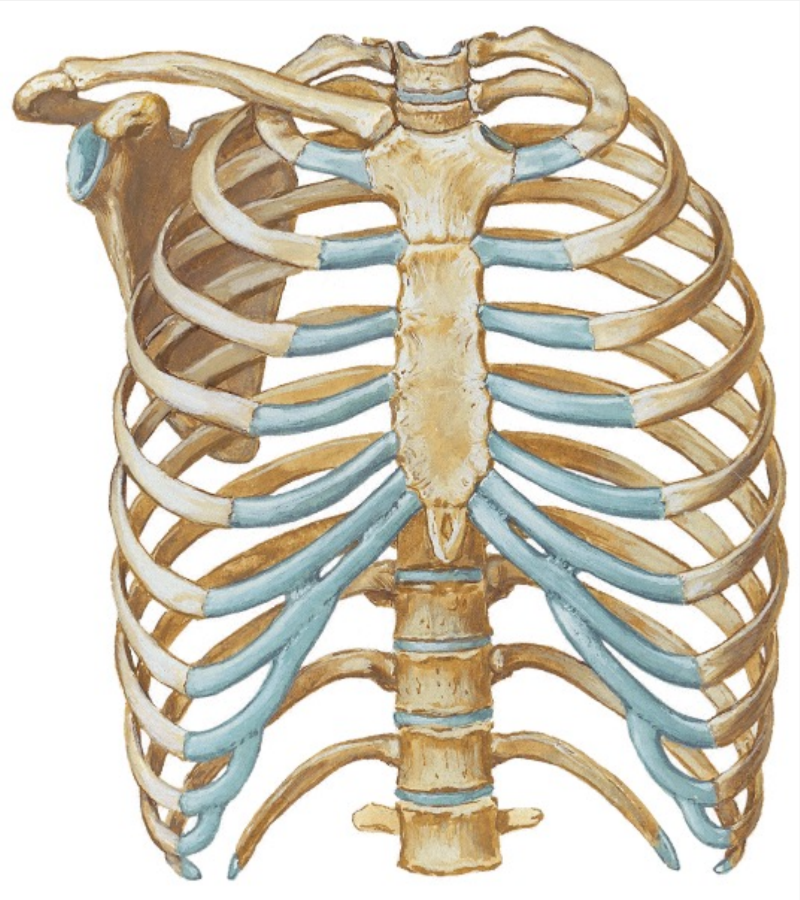
- **Made up of:**

1- Manubrium

2- Body

3- Xiphoid process

Sternal Biopsy:



Applied Notes

- Since the sternum possesses red hematopoietic marrow throughout life, it is a common site for **marrow biopsy**.
-
- The sternum may also be split **(median sternotomy)** at operation to allow the surgeon to gain easy access to the heart, great vessels, and thymus.